NATIONWIDE 5G STANDALONE (SA) FWA EXPANSION PROGRAM

DETAILED DESIGN DRAWING (DDD) DOSSIER • 30M HEAVY-DUTY POLYGONAL MONOPOLE

PROJECT TECHNICAL BASELINE & DESIGN PARAMETERS

TOWER CLASSIFICATION:	30.0M Heavy-Duty Polygonal Tubular Steel M
STRUCTURAL GEOMETRY:	5 Tapered Slip-Joint Sections (6,000mm len
DESIGN WIND VELOCITY:	Zone 1: 315 km/h (87.5 m/s) • Zone 2: 270
GOVERNING CODES:	NSCP 2015 (7th Ed.) Sec 207 • TIA-222-H •
SOIL BEARING CAPACITY:	qa >= 150 kPa (Standard) • qa >= 100 kPa (
FOUNDATION TYPE:	Pad & Chimney Mat: 5.0m x 5.0m x 0.8m Mat,
CONCRETE SPECIFICATION:	f'c >= 28 MPa (4,000 psi) @ 28 Days • Lean
REINFORCING STEEL:	ASTM A615 Grade 60 (Fy = 414 MPa) Deformed
STEEL SHAFT MATERIAL:	ASTM A572 Grade 50 (Fy = 345 MPa, Fu = 450
GALVANIZING STANDARD:	ASTM A123 / ISO 1461 Hot-Dip Galvanized >=
ANCHOR BOLT ASSEMBLY:	12x M36 Grade 105 (ASTM F1554) Anchor Bolt
BASEPLATE SPECIFICATION:	1,400mm OD x 60mm Thk Gr. 50 Plate with 12
TELECOM LOAD RATING:	3x Band n50 Cross-Pol Antennas (30m) + 3x
FUTURE EXPANSION:	1x 1.2m Solid Microwave Dish Mount @ 25m A
DEFLECTION LIMIT STATE:	Max Tip Sway <= H/100 (300mm) • Max Antenn
COMPOUND FOOTPRINT:	8.0m x 8.0m (64.0 m2) with 2.4m Cyclone Me
EQUIPMENT PLINTH:	2.0m x 2.0m x 0.3m Elevated Reinforced Con
POWER PLANT CAPACITY:	12kW Modular Rectifier (-48V DC) + 2x 48V
LIGHTNING PROTECTION:	16mm dia. x 1.5m Air Terminal (+31.5m AGL)
GROUNDING RESISTANCE:	Cadweld Exothermic Buried Ring Grid <= 5.0

MASTER DRAWING INDEX (SHEET REGISTER 01 - 30)

DWG NO.	SHEET TITLE	SCALE	REV
TT_AR_000	COVER PAGE & MASTER DRAWING INDEX	1:1	R0
TT_AR_001	DESIGN PARAMETERS, MATERIAL SPECS & 3D P	AS SHOWN	R0
TT_AR_002	PHILIPPINE BASIC WIND LOAD REGIMES & STR	NTS	R0
TT_AR_003	SITE LOCATION MAP, SEARCH RING & SITE DE	1:500	R0
TT_AR_004	STANDARD 8.0M x 8.0M COMPOUND PHYSICAL L	1:100	R0
TT_AR_005	30-METER MONOPOLE TOWER & COMPOUND ELEVA	1:100	R0
TT_ST_001	MONOPOLE KEY PLAN, SECTIONAL GEOMETRY &	1:50	R0
TT_ST_002	PAD AND CHIMNEY MAT FOUNDATION STRUCTURA	1:50	R0
TT_ST_003	FOUNDATION REINFORCEMENT & REBAR SECTION	1:25	R0
TT_ST_004	FOUNDATION REBAR BENDING SCHEDULE & MATE	NTS	R0
TT_ST_005	BASEPLATE TEMPLATE & HOLD-DOWN ANCHOR BO	1:20	R0
TT_ST_006	GENERAL CONCRETE SPECIFICATIONS & CIVIL	NTS	R0
TT_ST_007	STRUCTURAL STEEL FABRICATION, WELDING &	NTS	R0
TT_ST_008	OUTDOOR EQUIPMENT PAD PLINTH DETAILS (2.	1:25	R0
TT_ST_010	ELEVATED CABLE GANTRY & CABLE LADDER STR	1:20	R0

MASTER DRAWING INDEX CONTINUED (SHEETS 16 - 30):

DWG NO.	SHEET TITLE	SCALE	REV
TT_ST_011	COMPOUND SECURITY FENCING DETAILS (CYCLO	1:30	R0
TT_ST_012	COMPOUND ENTRANCE SWING GATE & SECURITY	1:25	R0
TT_ST_013	FENCE CORNER POST STIFFENERS & REINFORCE	1:20	R0
TT_ST_014	SERVICE ENTRANCE PEDESTAL & 230V UTILITY	1:25	R0
TT_ST_015	PERIMETER SECURITY LIGHTING & FIRE EXTIN	1:20	R0
TT_ST_017	BAND n50 CROSS-POL MACRO ANTENNA MOUNTIN	1:20	R0
TT_ST_018	STARLINK SATELLITE (10M AGL) & FUTURE EX	1:25	R0
TT_EL_001	ELECTRICAL SINGLE LINE DIAGRAM (SLD) & P	NTS	R0
TT_EL_002	SITE ELECTRICAL LAYOUT, TRENCHING & COND	1:100	R0
TT_EL_003	SITE EARTHING & GROUNDING GRID LAYOUT PL	1:100	R0
TT_EL_004	MAIN GROUNDING BUSBAR (MGB) & EARTH PIT	1:10	R0
TT_EL_005	GROUND TEST PIT & INSPECTION CHAMBER DET	1:15	R0
TT_EL_006	TOWER LIGHTNING DOWN-CONDUCTOR & CADWELD	1:20	R0
TT_EL_008	FIBER OPTIC LEAD-IN, DUCTING & UNDERGROU	1:20	R0
TT_ST_019	GEOTECHNICAL SOIL INVESTIGATION & BEARIN	NTS	R0

TENDER TYPICAL STATUS & ENGINEERING GOVERNANCE

1. DRAWING ISSUE STATUS:

- Issue Status: FOR TENDER / TYPICAL (Revision R0 / 26-OCT-2026).
- Represents nationwide typical design for 30M heavy-duty monopoles.
- Designed for 315 km/h Category 5 survivability & 150 kPa soil bearing.

2. MANDATORY CONDITIONS FOR AFC ISSUANCE:

Prior to Issued-For-Construction (IFC/AFC) status on any candidate site:

a. Confirmatory Soil Boring: Min 1 geotechnical borehole (SPT) to 10m.
b. Boundary Survey: Cadastral boundary & coordinates (WGS84 datum).
c. Site Wind Speed: Verify NSCP wind zone (Zone 3 floor: 250 km/h).
d. PE Wet-Seal: Review & authentication by licensed Structural PE.
e. Statutory Clearances: LGU building permit, Barangay & CAAP clearance.

3. ANTENNA & ACTIVE RADIO EQUIPMENT LOADING:

- Primary Telecom: 3x 5G Band n50 Cross-Pol Antennas @ 30.0m AGL.
- Active Radios: 3x Band n50 8T8R Remote Radio Units (38 kg each).
- Transmission: 1x Starlink High-Performance dish @ 10.0m AGL.
- Future Allowance: 1x 1.2m solid microwave dish mount @ 25.0m AGL.

4. GOVERNING MATERIAL SPECIFICATIONS:

- Shaft Steel: ASTM A572 Grade 50 (Fy = 345 MPa / 50 ksi).
- Hot-Dip Galvanizing: ASTM A123 / ISO 1461 >= 85 microns zinc.
- Anchor Bolts: 12x M36 Grade 105 (ASTM F1554) on 1,200mm PCD.
- Concrete: f'c >= 28 MPa (4,000 psi) @ 28 days (sulfate-resistant).
- Earthing Grid: Cadweld exothermically welded copper ring <= 5.0 Ohms.

CRITICAL TENDER COMPLIANCE NOTICE:

This DDD package controls standard structural geometry and BoQ billable scope. In the event of discrepancy with legacy reference sketches, this 30-sheet DDD manual and Appendix D govern all technical evaluations.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — **TELTECH 5G FWA**

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

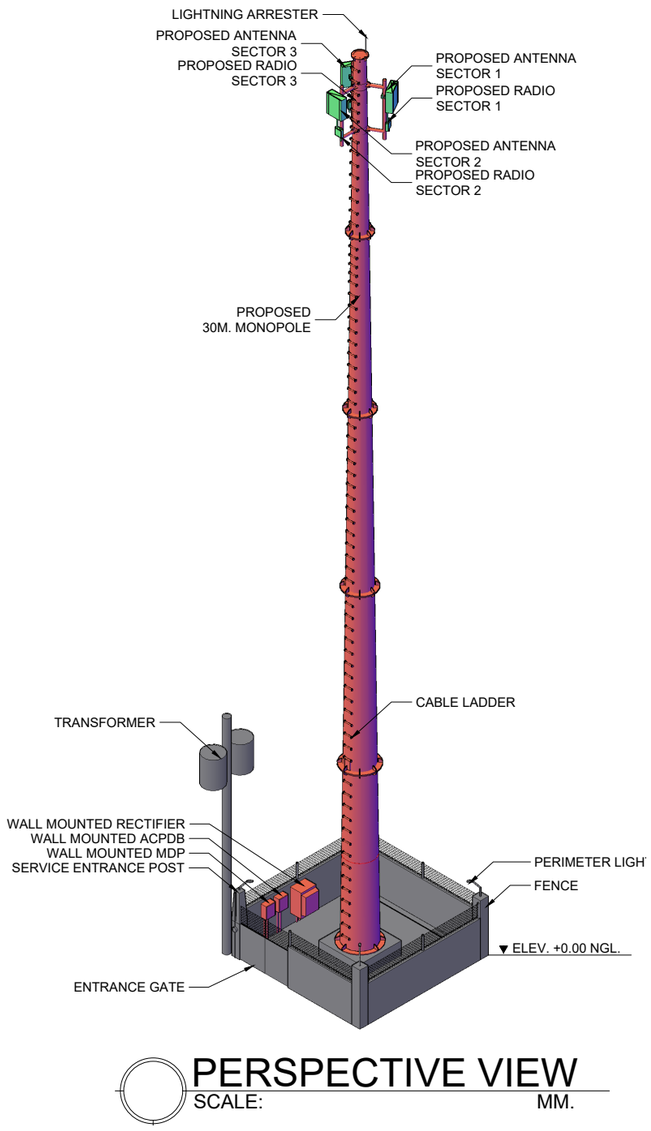
SHEET TITLE: COVER PAGE & MASTER DRAWING INDEX

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_AR_000	FOR TENDER / TYPICAL	R0 / 26-OCT-26	01 OF 30

30-METER MONOPOLE GENERAL SPECIFICATIONS & 3D PERSPECTIVE

STRUCTURAL DESIGN CRITERIA, MATERIAL STANDARDS & APPARENT ISOMETRIC LAYOUT

TOWER & SITE 3D PERSPECTIVE VIEW (AUTOCAD VECTOR DWG)



PERSPECTIVE VIEW
SCALE: MM.

CAD PERSPECTIVE VIEW EMBEDDED FROM ARCHITECTURAL VECTOR MASTER DWG

STRUCTURAL DESIGN CRITERIA & MATERIAL SPECIFICATION

1. APPLICABLE DESIGN CODES & GOVERNING SPECIFICATIONS:

STANDARD / CODE	ISSUING BODY	APPLICATION SCOPE
NSCP 2015 (7th Ed.)	ASEP / DPWH	Philippine Wind Loading & Seismic Design (Sec. 207 & 208)
ANSI/TIA-222-H	TIA / EIA	Structural Standard for Antenna Supporting Structures & Antennas
AISC 360-16 / 341-16	AISC	Specification for Structural Steel Buildings & Seismic Provisions
ACI 318-19	ACI	Building Code Requirements for Structural Concrete & Foundation Mat
AWS D1.1/D1.1M:2020	AWS	Structural Welding Code - Steel (Full Penetration & Fillet Welds)
ASTM A123 / ISO 1461	ASTM / ISO	Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
PEC 2017 (Part 1)	IIEE / PRC	Philippine Electrical Code: Grounding, Lightning & Overcurrent

2. MONOPOLE SECTIONAL GEOMETRY & WALL THICKNESSES:

SEC	ELEV. RANGE (m)	LENGTH	BASE OD	TOP OD	THK	STEEL GRADE	WEIGHT
S1	0.00m - 6.00m	6,000mm	1,050mm	960mm	14mm	ASTM A572 Gr 50	2,240 kg
S2	4.50m - 10.50m	6,000mm	982mm	892mm	12mm	ASTM A572 Gr 50	1,780 kg
S3	9.00m - 15.00m	6,000mm	914mm	824mm	12mm	ASTM A572 Gr 50	1,650 kg
S4	13.50m - 19.50m	6,000mm	846mm	756mm	8mm	ASTM A572 Gr 50	1,020 kg
S5	18.00m - 30.00m	12,000mm	778mm	450mm	6mm	ASTM A572 Gr 50	1,180 kg

3. TELECOMMUNICATIONS EQUIPMENT LOADING SCHEDULE:

ITEM DESCRIPTION	QTY	ELEVATION	DIMENSIONS (mm)	WEIGHT (EA)	EPA (m2)
Band n50 Cross-Pol Macro Antenna	3	+30.0m AGL	1400 x 350 x 160	28.0 kg	0.68 m2
Band n50 8T8R Remote Radio Unit (RRU)	3	+29.0m AGL	480 x 360 x 180	38.0 kg	0.22 m2
Starlink High-Performance Satellite Dish	1	+10.0m AGL	575 x 511 x 50	7.0 kg	0.32 m2
Future 1.2m Solid Microwave Backhaul	1	+25.0m AGL	1200 dia. x 450	45.0 kg	1.25 m2
Lightning Air Terminal (Copper/SS)	1	+31.5m AGL	16 dia. x 1500	4.5 kg	0.03 m2
Safety Climb Cable & Sleeve Assembly	1	0m - 30.0m	8mm dia. Galvanized	25.0 kg	Line

4. CRITICAL STRUCTURAL MANDATES & TOLERANCES:

- Slip joints shall have a minimum overlap length of 1.5 times the internal sleeve diameter (1,500mm standard).
- All pole shaft longitudinal welds shall achieve 100% full penetration in the male/female slip joint lap zones.
- Tension bolts and anchor assemblies must be torqued using calibrated tension wrenches in star pattern.
- No field welding, hole torching, or field alterations permitted without written approval from Structural Engineer.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — **TELTECH 5G FWA**

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

SHEET TITLE: DESIGN PARAMETERS, MATERIAL SPECS & 3D PERSPECTI

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_AR_001	FOR TENDER / TYPICAL	R0 / 26-OCT-26	02 OF 30

NSCP 2015 SECTION 207 WIND LOAD ZONING & STRUCTURAL LIMIT STATES

SURVIVABILITY & OPERATIONAL DEFLECTION CRITERIA FOR EXTREME TYPHOON RESILIENCE

PHILIPPINE BASIC WIND SPEED MAP (NSCP 2015 7TH ED. FIGURE 207A)

PAGASA / NSCP TYPHOON EXPOSURE PROFILES:

ZONE 1 (EXTREME EXPOSURE): **V = 315 km/h (87.5 m/s) • 3-sec gust, 50-yr MRI**

Coverage: Eastern Luzon, Bicol, Samar, Leyte, Eastern Mindanao

ZONE 2 (SEVERE EXPOSURE): **V = 270 km/h (75.0 m/s) • 3-sec gust, 50-yr MRI**

Coverage: Central Visayas, Western Luzon, Panay, Negros, Cebu

ZONE 3 (PROC

Coverage: Pa

NATIONWIDE 5G STANDALONE (SA) F

SHEET 2: GENERAL STRUCTURAL DESIGN

1. PHILIPPINE BASIC WIND SPEED REGIMES (NSCP 2015 / PAGASA HISTORICAL TYPHOON RECURRENC

Philippine Wind Zone (NSCP 2015 7th Ed. Sec. 207)	Ultimate Basic Wind Speed (V) (3-sec gust, 50-yr return)	Nominal Wind Speed (Vasd)
ZONE 1 (Typhoon Track / Eastern Seaboard)	315 km/h (87.5 m/s)	245 km/h (68.0 m/s)
ZONE 2 (Central/West Luzon, Visayas)	270 km/h (75.0 m/s)	210 km/h (58.3 m/s)
ZONE 3 (Inland Mindanao & Palawan)	200 km/h (55.6 m/s)	155 km/h (43.1 m/s)

2. TOWER OPERATIONAL & SURVIVABLE WIND LIMIT STATES AND DEFLECTION CRITERIA (TIA-222-H /

Survivable Wind Speed (Vsurvival)	**270 km/h to 315 km/h (3-sec gust, Ultimate Limit State / LRFD)**	Full str
Operational Wind Speed (Voperational)	**140 km/h to 160 km/h (40.0 m/s to 44.4 m/s, Serviceability Limit State)**	Structu
Allowable Twist (θtwist / Torsional Rotation)	**≤ ±0.5° (30 arcmin) for MW / Backhaul ≤ ±1.0° (60 arcmin) for 5G Macro Sector**	Maximu
Allowable Sway / Tilt (θtilt)	**≤ ±1.0° (60 arcmin) at top antenna mounting centerline (30.0m AGL)**	Angula
Allowable Verticality (Tower Plumbness)	**≤ 1 : 500 (0.20% of Tower Height H) (Max 60 mm for 30m tower; 80 mm for 40m tower)**	Out-of-
Top Horizontal Service Deflection	**≤ H / 100 (Max 300 mm at 30.0m AGL)**	Maximu

3. CIVIL, STRUCTURAL & ELECTRICAL MATERIAL SPECIFICATIONS

Item / Component	Standard Engineering Specification
Monopole Shaft Steel	ASTM A572 Grade 50 (Fy = 345 MPa / 50 ksi, Fu = 450 MPa). 12/18-sided polygon.
Slip-Joint Overlap	Minimum 1.5 x inside mating diameter per section joint.
Corrosion Protection	Hot-dip galvanized per ASTM A123 / ISO 1461 (≥ 85 µm zinc, ≥ 610 g/m²).
Compound Footprint	8.0m x 8.0m (64 m²) gravel-surfaced compound with geotextile barrier.

OPERATOR / APPLICANT: Telecommunications Technology Solutions Inc. **NETWORK PARTNER:** Comclark Network and Technology Corp.	**ENGINEER:** JOMAR DOGILLO PRC No.: _____ PTR No.: _____	**AUTHORIZE OLIVER TUN** Lead Technic

WIND PRESSURE FORMULATION (NSCP 2015 SEC. 207 & TIA-222-H)

1. VELOCITY PRESSURE EQUATION:

$$qz = 0.613 * Kz * Kzt * Kd * Ke * V^2 * Iw \quad (N/m2)$$

- **qz** : Velocity pressure at height z above ground level (N/m2)
- **Kz** : Velocity pressure exposure coefficient = 2.01 * (z / zg)^(2 / alpha) = 1.18 @ 30m
- **Kzt** : Topographic factor = [1 + K1*K2*K3]^2 = 1.0 (Flat / Rolling Terrain)
- **Kd** : Wind directionality factor = 0.95 (Tubular Polygonal Pole Structure)
- **Ke** : Ground elevation factor = 1.00 (Sea level to 500m ASL nominal)
- **V** : Basic wind speed (3-second gust) = 87.5 m/s (315 km/h for Zone 1)
- **Iw** : Importance factor for telecommunications infrastructure = 1.15 (Essential Facility)

2. DESIGN VELOCITY PRESSURE GRADIENT (ZONE 1: 315 KPH):

HEIGHT z (m)	Kz (EXP C)	V (m/s)	qz (N/m2)	qz (kPa)	APPLIED FORCE / M
0.0m - 6.0m	0.85	87.5 m/s	4,480 N/m2	4.48 kPa	4.70 kN/m
6.0m - 12.0m	0.98	87.5 m/s	5,165 N/m2	5.17 kPa	4.85 kN/m
12.0m - 18.0m	1.06	87.5 m/s	5,587 N/m2	5.59 kPa	4.64 kN/m
18.0m - 24.0m	1.13	87.5 m/s	5,956 N/m2	5.96 kPa	4.17 kN/m
24.0m - 30.0m	1.18	87.5 m/s	6,219 N/m2	6.22 kPa	3.42 kN/m

STRUCTURAL LIMIT STATES & SERVICEABILITY DEFLECTION CRITERIA

LIMIT STATE	WIND SPEED	CRITERIA / REQUIREMENT	CALCULATED	STATUS
Serviceability Sway	180 km/h (Op)	Delta_tip <= H / 100 = 300mm	185 mm	PASS (SF = 1.62)
Serviceability Tilt	180 km/h (Op)	Theta <= 1.00 deg @ Antenna CL	0.62 deg	PASS (SF = 1.61)
Ultimate Flexure (Mu)	315 km/h (Ult)	Mu <= Phi * Mn (Grade 50)	1,850 kNm	PASS (UR = 0.84)
Ultimate Shear (Vu)	315 km/h (Ult)	Vu <= Phi * Vn (Shaft Base)	145 kN	PASS (UR = 0.42)
Foundation Bearing	315 km/h (Ult)	q_max <= qa = 150 kPa	118 kPa	PASS (SF = 2.45)
Overturning (SF)	315 km/h (Ult)	SF_ot >= 1.50 (Resisting/OT)	2.15	PASS (Resilient)
Sliding Resistance	315 km/h (Ult)	SF_sl >= 1.50 (Friction + Pass)	1.85	PASS (Stable)

3. STRUCTURAL SAFETY GOVERNANCE NOTES:

- P-Delta secondary moments are rigorously incorporated in all finite-element sectional analyses.
- Torsional twist at the antenna centerline is constrained below 0.35 degrees under operational wind.
- Vibration mitigation: Internal damping cables installed if vortex shedding occurs in 15-25 km/h wind regime.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — **TELTECH 5G FWA**

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SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

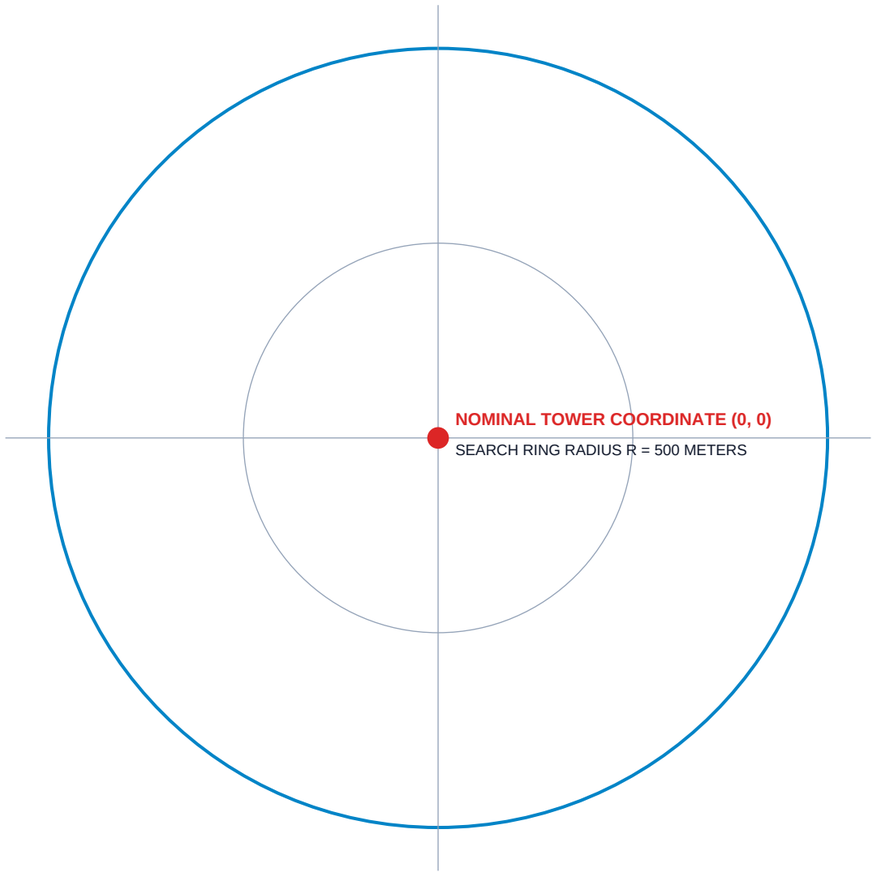
SHEET TITLE: PHILIPPINE BASIC WIND LOAD REGIMES & STRUCTURAL

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TT_AR_002	FOR TENDER / TYPICAL	R0 / 26-OCT-26	03 OF 30

RADIO ACCESS NETWORK (RAN) NOMINAL SITING & ACCESS CORRIDOR

CANDIDATE SEARCH RING (R=500M), RIGHT-OF-WAY (ROW) & ENVIRONMENTAL SETBACK MATRIX

CANDIDATE SEARCH RING & VICINITY TOPOGRAPHY (SCALE 1:500)



SITE ACQUISITION (SAQ) & CIVIL PLANNING CRITERIA

1. GEOGRAPHIC & ENVIRONMENTAL SITING PARAMETERS:

PARAMETER	PROJECT SPECIFICATION	ENGINEERING JUSTIFICATION
Geodetic Datum	WGS 84 / PRS 92 Projection	Direct compatibility with NTC GIS portal
Search Ring Radius	500.0 Meters (Rural GIDA)	Maintains RF coverage & link budget integrity
Terrain Classification	Exposure Category C (Open)	Rolling grasslands, shoreline, rural agricultural
Topographic Factor (Kzt)	1.00 (Flat) to 1.35 (Escarpment)	Field verified prior to foundation casting
Access Road (ROW)	Minimum 3.0m Clear Width	Crane & concrete transit mixer accessibility
Utility Power Lead-in	230V 1-Ph / 3-Ph 60Hz Drop	Within 150m of distribution line tap pole

2. MINIMUM PROPERTY SETBACK REQUIREMENTS (DPWH / LGU):

BOUNDARY	REQUIRED SETBACK	TELTECH DESIGN	COMPLIANCE STATUS
Front (Access Road / Gate)	Minimum 3.00 Meters	4.50 Meters	COMPLIANT (+1.50m margin)
Rear Property Line	Minimum 2.00 Meters	3.00 Meters	COMPLIANT (+1.00m margin)
Left Lateral Boundary	Minimum 2.00 Meters	2.50 Meters	COMPLIANT (+0.50m margin)
Right Lateral Boundary	Minimum 2.00 Meters	2.50 Meters	COMPLIANT (+0.50m margin)
High-Voltage Power Lines	Minimum 15.00 Meters	20.00 Meters	COMPLIANT (Safety clearance)

3. SITE SELECTION & CIVIL CLEARANCE MANDATES:

- Candidate sites must be situated outside 100-year flood hazard inundation zones per MGB/NOAH maps.
- Soil bearing capacity must be experimentally verified via two (2) standard penetration test (SPT) boreholes.
- Sites located within 15 km of coastal waters must implement marine-grade heavy galvanizing (>= 100 microns).
- Compound finished grade level (FGL) shall be established at minimum +300mm above natural ground level (NGL).
- A continuous perimeter concrete grade curb (200mm x 300mm) must encircle the entire 8m x 8m compound.

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OLIVER TUNGOL	JOMAR DOGILLO, PE

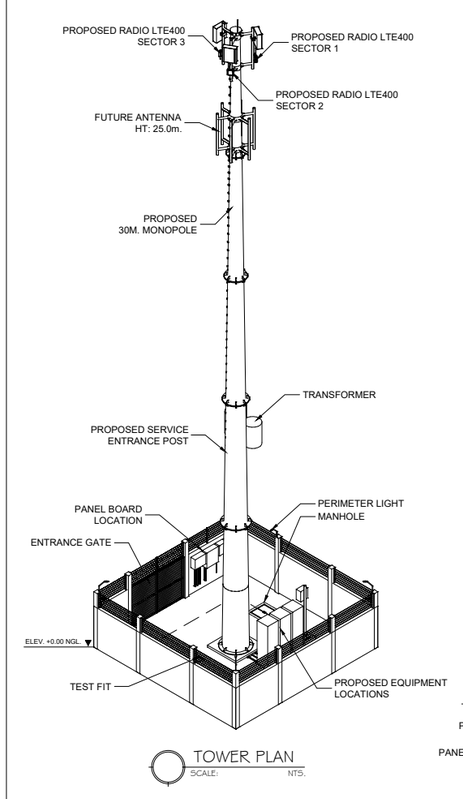
SHEET TITLE: SITE LOCATION MAP, SEARCH RING & SITE DEVELOPMEN

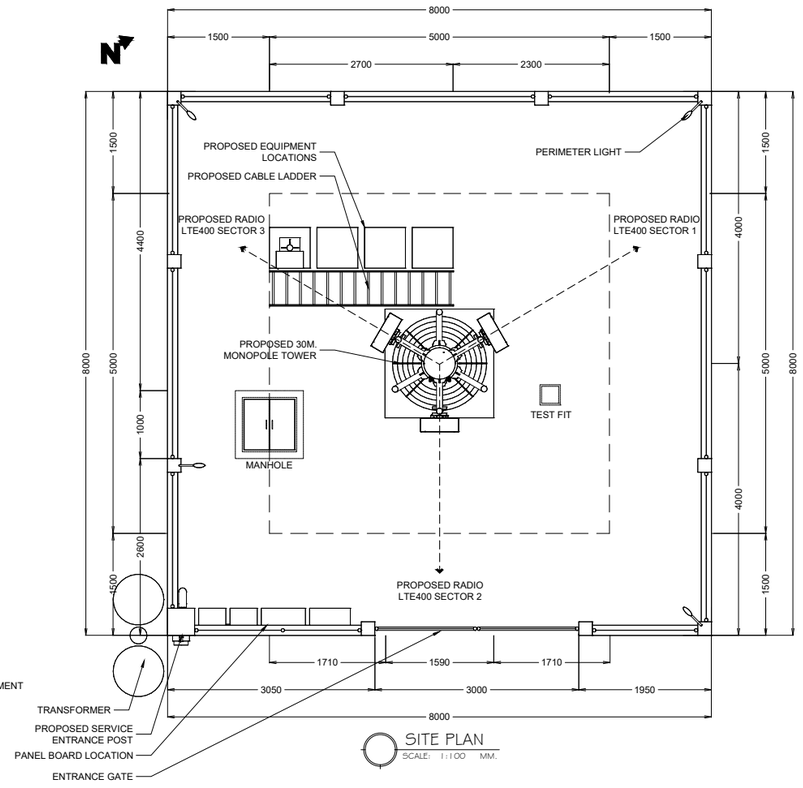
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TT_AR_003	FOR TENDER / TYPICAL	R0 / 26-OCT-26	04 OF 30

8.0M x 8.0M COMPOUND MASTER PHYSICAL LAYOUT & EQUIPMENT PLACEMENT

FOOTPRINT COORDINATION • MONOPOLE MAT • EQUIPMENT PLINTH • CABLE BRIDGE • EARTHING

STANDARD 8.0M x 8.0M COMPOUND PLAN (SCALE 1:100)





OPERATOR: Telecommunications Technology Solutions Inc	OLIVER TUNGOL ENGINEER	SITE PLAN			AUTHORIZED REPRESENTATIVE:	NO.	DATE	DESCRIPTION	SHEET CONTENT: / SHEET NO.:
	PRC NO.: / TIN NO.:								
	DATE ISSUED: / PTR NO.:	ADDRESS:		DRAWN / CHECKD / APPRVD					DRAWING BY: JOMAR DOGILLO / REV
	PLACE ISSUED:	LONGITUDE:	LATITUDE:						

CAD PHYSICAL LAYOUT PLAN EMBEDDED FROM ARCHITECTURAL VECTOR MASTER DWG

COMPOUND COORDINATES & COMPONENT SCHEDULE

1. FACILITY COORDINATE MATRIX (ORIGIN @ TOWER CL):

FACILITY COMPONENT	X (m)	Y (m)	SIZE (m)	ELEVATION
Monopole Tower Base	0.00	0.00	1.05m dia.	+0.30m AGL
Foundation Mat Footing	0.00	0.00	5.0m x 5.0m	-1.70m Founding
Outdoor Equipment Pad	+2.20	-1.50	2.0m x 2.0m	+0.30m AGL
Overhead Cable Gantry	+1.10	-0.75	0.4m x 3.5m	+2.50m Clearance
Service Entrance Post	+3.80	-3.50	0.15m x 0.15m	+3.50m Top
Main Entrance Gate (1.2m)	-1.50	-4.00	1.2m x 2.4m	Compound Level
Earth Test Pit 1 (NE)	+3.60	+3.60	0.3m x 0.3m	Flush with Grade
Earth Test Pit 2 (NW)	-3.60	+3.60	0.3m x 0.3m	Flush with Grade
Earth Test Pit 3 (SW)	-3.60	-3.60	0.3m x 0.3m	Flush with Grade
Earth Test Pit 4 (SE)	+3.60	-3.60	0.3m x 0.3m	Flush with Grade

2. COMPOUND CONSTRUCTION SCHEDULE:

CIVIL ITEM	QUANTITY	SPECIFICATION / MATERIAL
Total Compound Area	64.0 m2	8.0m x 8.0m boundary inside fence line
Crushed Gravel Bedding	6.4 m3	100mm layer G-1 crushed stone (weed barrier)
Reinforced Grade Curb	32.0 Linear m	200mm W x 300mm D with 4-12mm rebar
Cyclone Perimeter Fence	32.0 Linear m	2.4m H No. 10 gauge mesh + 0.5m Y-arms
Entrance Swing Gate	1 Set	1.2m W x 2.4m H with slide bolt & lock box
Underground Power Conduit	18.0 Linear m	50mm dia. Schedule 40 PVC concrete encased
Perimeter Lighting Fixtures	2 Sets	50W LED Floodlights on opposite diagonal posts

3. COMPOUND DRAINAGE & CIVIL MANDATES:

- Finished grade within compound shall slope outward at 1.5% minimum towards boundary curb weep holes.
- Heavy-duty geotextile weed barrier fabric shall be installed beneath the 100mm gravel surfacing.
- Underground conduits must maintain 300mm minimum separation between AC power and telecommunications fiber.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — **TELTECH 5G FWA**

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

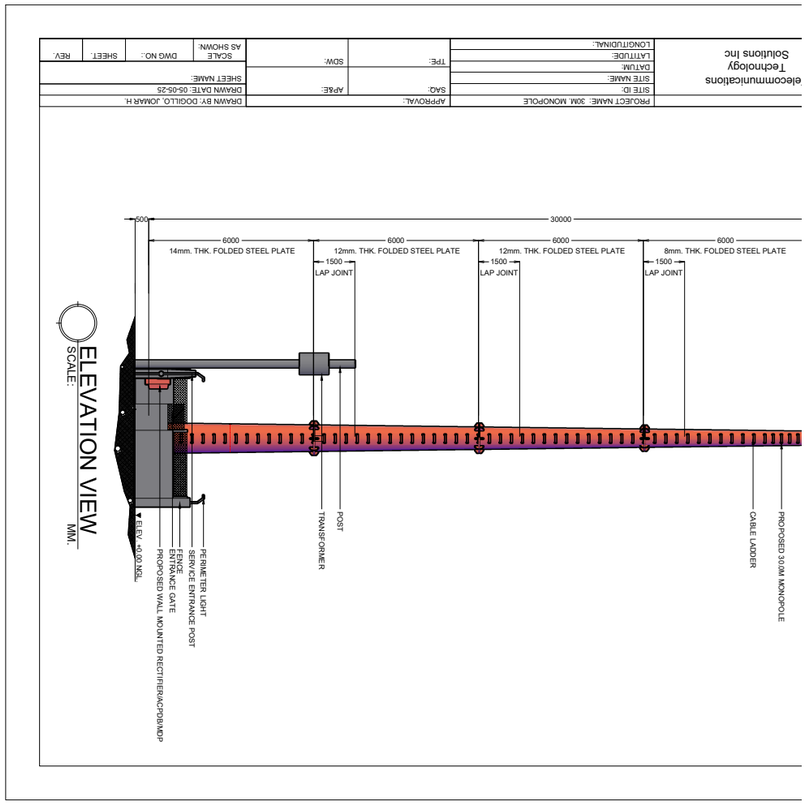
SHEET TITLE: STANDARD 8.0M x 8.0M COMPOUND PHYSICAL LAYOUT PL

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_AR_004	FOR TENDER / TYPICAL	R0 / 26-OCT-26	05 OF 30

30-METER MONOPOLE TOWER ELEVATION & APPURTENANCE STACKING

VERTICAL PROFILE • SLIP-JOINT TRANSITIONS • ANTENNA AZIMUTHS • POWER & FIBER ROUTING

30-METER MONOPOLE VERTICAL ELEVATION PROFILE (SCALE 1:100)



CAD ELEVATION VIEW EMBEDDED FROM ARCHITECTURAL VECTOR MASTER DWG

VERTICAL ELEVATION SCHEDULE & SECTIONAL STACKING

1. VERTICAL DATUMS & APPURTENANCE ELEVATIONS:

DATUM / APPURTENANCE	AGL (m)	QTY	MOUNTING / DETAILS
Lightning Air Terminal	+31.50m	1 Set	16mm dia. SS/Copper rod at tower peak
Band n50 Macro Sector Antennas	+30.00m	3 Sets	Cross-Pol panels @ 0, 120, 240 deg
Band n50 8T8R RRU Units	+29.00m	3 Sets	Directly behind antennas on mount pipes
Future Microwave Dish Mount	+25.00m	1 Set	1.2m solid dish outrigger collar mount
Starlink High-Performance Terminal	+10.00m	1 Set	1.2m outrigger pipe mount (open sky)
Overhead Cable Gantry Bridge	+2.50m	1 Span	Clearance across compound access path
Compound Security Fence & Razor	+2.90m	Perimeter	2.4m cyclone + 0.5m concertina wire
Pedestal Chimney Top	+0.30m	1 Mat	Baseplate bedding (+50mm grout bed)
Natural Ground Level (NGL)	0.00m	Reference	Finished compound surface (+0.00m)
Foundation Founding Level	-1.70m	Base	Blinding bed level on competent soil

2. SLIP-JOINT STACKING TOLERANCES:

JOINT ID	ELEVATION	LAP LENGTH	MALE SEC	FEMALE SEC
Slip Joint J1-2	+4.50m to +6.00m	1,500 mm	Section S2	Section S1 (Base)
Slip Joint J2-3	+9.00m to +10.50m	1,500 mm	Section S3	Section S2
Slip Joint J3-4	+13.50m to +15.00m	1,500 mm	Section S4	Section S3
Slip Joint J4-5	+18.00m to +19.50m	1,500 mm	Section S5	Section S4 (Top)

3. ERECTION & FIELD ASSEMBLY MANDATES:

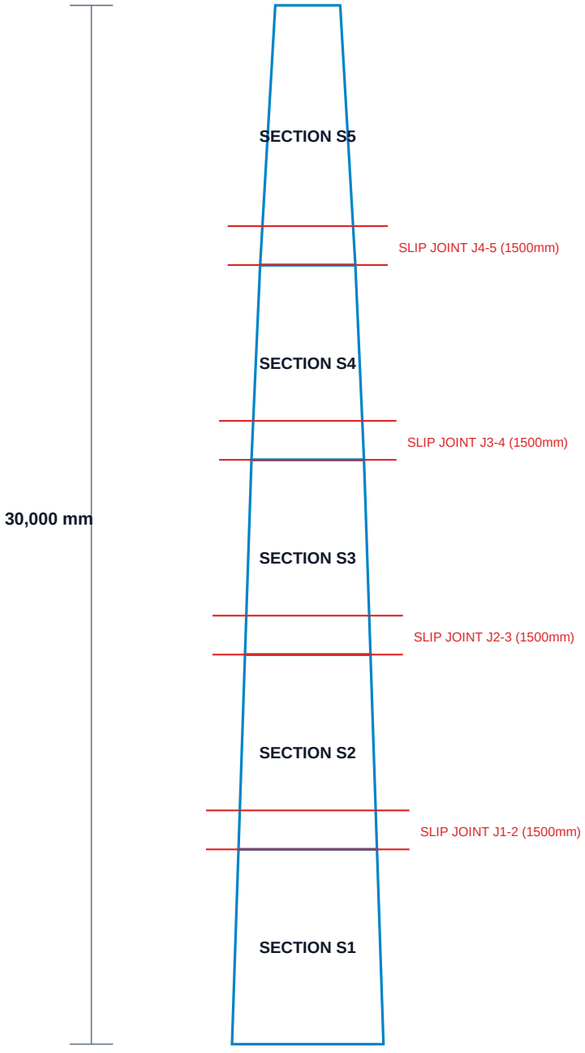
- Monopole sections must be pulled to full match-mark engagement using two 10-ton hydraulic jacks.
- Total out-of-plumb tolerance along entire 30m height shall not exceed 1:500 (60mm maximum tip offset).
- All RF and power feeder cables must run internally within the hollow polygonal pole shaft.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC.	TELTECH 5G FWA
PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50) SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa	
LEAD TECHNICAL ARCHITECT / REP: OLIVER TUNGOL	PROFESSIONAL STRUCTURAL ENGINEER: JOMAR DOGILLO, PE
SHEET TITLE: 30-METER MONOPOLE TOWER & COMPOUND ELEVATION VIE	
DRAWING NO: TT_AR_005	RELEASE STATUS: FOR TENDER / TYPICAL
REV / DATE: R0 / 26-OCT-26	SHEET NO: 06 OF 30

MONOPOLE SHAFT GEOMETRY, SECTION PROPERTIES & DESIGN REACTIONS

12-SIDED POLYGONAL SHAFT • TAPER RATIO (15 MM/M) • ULTIMATE BASE REACTIONS (315 KPH)

MONOPOLE SECTIONAL STACKING & TAPER PROFILE (SCALE 1:50)



SECTION PROPERTIES & CRITICAL STRUCTURAL REACTIONS

1. FINITE ELEMENT SECTION PROPERTY MATRIX:

SEC	LENGTH	BASE DIA	TOP DIA	THK	Ix (cm4)	Zx (cm3)	WT (kg)
S1	6,000mm	1,050mm	960mm	14mm	5,420,000	103,200	2,240 kg
S2	6,000mm	982mm	892mm	12mm	3,750,000	76,400	1,780 kg
S3	6,000mm	914mm	824mm	12mm	2,980,000	65,100	1,650 kg
S4	6,000mm	846mm	756mm	8mm	1,720,000	40,700	1,020 kg
S5	12,000mm	778mm	450mm	6mm	840,000	21,500	1,180 kg

2. FOUNDATION BASE REACTIONS (NSCP 2015 & TIA-222-H):

LOAD CASE / LIMIT STATE	MOMENT (kNm)	SHEAR (kN)	AXIAL (kN)	TORSION
ULS: 1.2D + 1.0W (315 km/h)	1,850 kNm	145 kN	120 kN	12.5 kNm
ULS: 0.9D + 1.0W (Max Uplift)	1,825 kNm	145 kN	85 kN	12.5 kNm
SLS: 1.0D + 1.0W (180 km/h)	600 kNm	48 kN	100 kN	4.2 kNm
Seismic: 1.2D + 1.0E (Zone 4)	420 kNm	38 kN	125 kN	1.8 kNm
Dead Load Only (Gravity)	0 kNm	0 kN	100 kN	0 kNm

3. STRUCTURAL UTILIZATION RATIO (UR) SUMMARY:

STRUCTURAL ELEMENT	DESIGN CAPACITY	APPLIED DEMAND	UR	STATUS
Pole Shaft Base Flexure	Phi*Mn = 2,205 kNm	Mu = 1,850 kNm	0.84	PASS (Safe)
Pole Shaft Base Shear	Phi*Vn = 345 kN	Vu = 145 kN	0.42	PASS (Safe)
Slip Joint J1-2 Lap Shear	Phi*Vn = 410 kN	Vu = 115 kN	0.28	PASS (Safe)
Baseplate Flexural Stress	Phi*Mp = 485 kNm	Mu = 390 kNm	0.80	PASS (Safe)
Anchor Bolt Tensile Pull	Phi*Tn = 620 kN/ea	Tu = 445 kN/ea	0.72	PASS (Safe)

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)

SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

SHEET TITLE: MONOPOLE KEY PLAN, SECTIONAL GEOMETRY & STRUCTUR

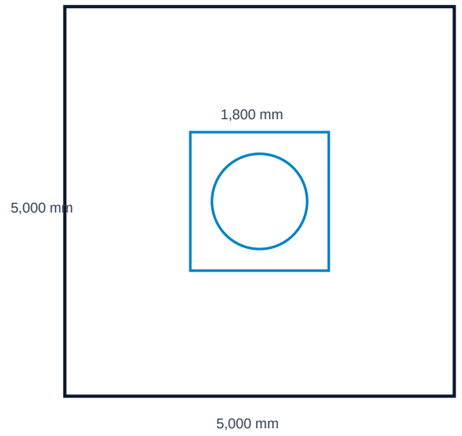
DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_ST_001	FOR TENDER / TYPICAL	R0 / 26-OCT-26	07 OF 30

REINFORCED CONCRETE PAD & CHIMNEY FOUNDATION MASTER DETAILS

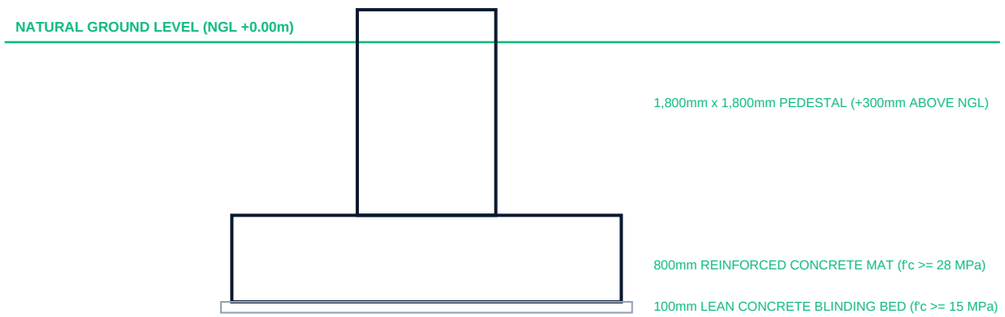
5.0M x 5.0M x 0.8M MAT • 1.8M x 1.8M PEDESTAL • BEARING CAPACITY (qa >= 150 KPA)

FOUNDATION STRUCTURAL PLAN & SECTION (SCALE 1:50)

FOUNDATION PLAN VIEW (5.0M x 5.0M MAT):



FOUNDATION CROSS SECTION VIEW:



GEOTECHNICAL STABILITY & BEARING CALCULATIONS

1. FOUNDATION GEOMETRIC PARAMETERS & WEIGHTS:

COMPONENT	DIMENSIONS	VOLUME	UNIT WT	TOTAL WEIGHT
RC Mat Footing	5.00m x 5.00m x 0.80m	20.00 m3	24.0 kN/m3	480.0 kN
RC Pedestal Chimney	1.80m x 1.80m x 1.20m	3.89 m3	24.0 kN/m3	93.4 kN
Compacted Soil Surcharge	Overburden (0.90m depth)	19.00 m3	18.0 kN/m3	342.0 kN
Monopole + Telecom Gear	30m Tower Shaft + Antennas	Steel	Gravity Dead	120.0 kN
TOTAL VERTICAL LOAD (W)	Total Gravity Foundation Load	--	--	1,035.4 kN

2. SOIL PRESSURE & STABILITY FACTOR CHECKS (qa >= 150 kPa):

CHECK PARAMETER	GOVERNING FORMULA	LIMIT VALUE	CALCULATED	STATUS
Eccentricity (e)	e = M_total / W_total	e <= B / 6 = 0.83m	0.78 m	PASS (Kern)
Max Soil Pressure (q_max)	q = (W/A) * [1 + 6e/B]	q <= 150.0 kPa	118.4 kPa	PASS (SF = 2.45)
Min Soil Pressure (q_min)	q = (W/A) * [1 - 6e/B]	q >= 0 (No Uplift)	12.2 kPa	PASS (Compression)
Overturning Stability	SF_ot = M_resist / M_ot	SF >= 1.50	2.15	PASS (+43% margin)
Sliding Resistance	SF_sl = (mu*W + Pp) / V	SF >= 1.50	1.85	PASS (+23% margin)

3. MANDATORY CIVIL CONSTRUCTION DIRECTIVES:

- Excavation subgrade must be inspected and approved by Geotechnical Engineer prior to blinding placement.
- A 100mm lean concrete blinding layer (f'c >= 15 MPa) must be placed immediately to seal competent subgrade.
- Backfill material over the foundation mat shall be clean granular soil compacted in 200mm layers to 95% MDD.
- No groundwater dewatering discharge shall erode adjacent footings or property boundaries.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — **TELTECH 5G FWA**

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

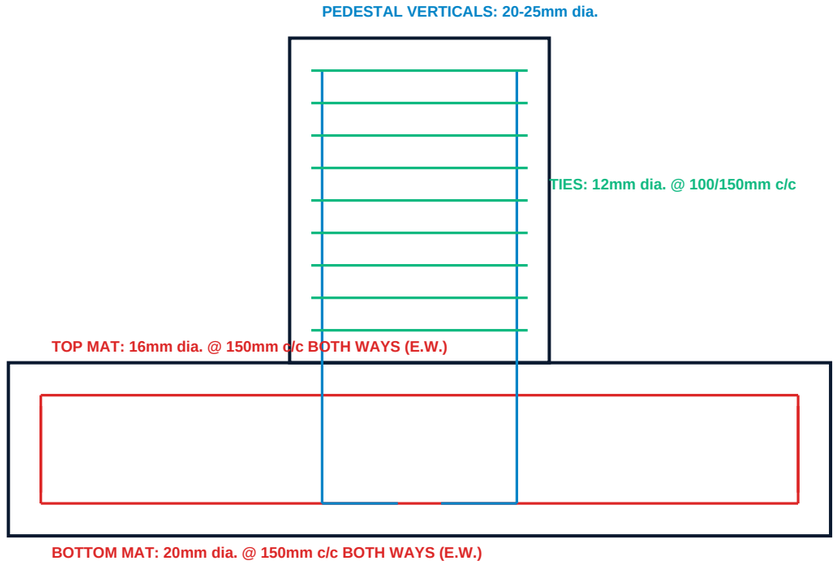
SHEET TITLE: PAD AND CHIMNEY MAT FOUNDATION STRUCTURAL DETAIL

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_ST_002	FOR TENDER / TYPICAL	R0 / 26-OCT-26	08 OF 30

FOUNDATION REINFORCEMENT DETAILS & REBAR PLACEMENT SCHEDULE

TOP & BOTTOM MAT GRIDS • PEDESTAL DOWELS • LATERAL TIES • CLEAR COVER (75MM / 50MM)

REINFORCING REBAR SECTIONAL DETAILS (SCALE 1:25)



REBAR SPECIFICATION & REINFORCEMENT SCHEDULE

1. REINFORCING STEEL DESIGN SPECIFICATIONS:

ELEMENT	REBAR DIA	SPACING / QTY	GRADE	MIN LAP LENGTH
Mat Bottom (E.W.)	20 mm dia.	@ 150 mm c/c	ASTM A615 Gr 60	900 mm (45 db)
Mat Top (E.W.)	16 mm dia.	@ 150 mm c/c	ASTM A615 Gr 60	720 mm (45 db)
Pedestal Verticals	25 mm dia.	20 Nos. Evenly Spaced	ASTM A615 Gr 60	1,125 mm (45 db)
Pedestal Lateral Ties	12 mm dia.	@ 100mm (top) / 150mm	ASTM A615 Gr 60	135 deg seismic hook
Foundation Dowels	25 mm dia.	20 Nos. Embedded to Mat	ASTM A615 Gr 60	Full development leg

2. MINIMUM CONCRETE COVER TO REBAR (ACI 318-19):

SURFACE / LOCATION	MIN CLEAR COVER	PURPOSE / CORROSION MITIGATION
Mat Bottom (Cast against earth)	75 mm	Protection against aggressive soil ions & groundwater
Mat Sides (Formed with soil backfill)	50 mm	Protection against moisture penetration
Mat Top Surface	50 mm	Weathering and shrinkage crack isolation
Pedestal Chimney Vertical Faces	50 mm	Atmospheric carbonation & weathering protection
Pedestal Top (Beneath Baseplate Grout)	50 mm	Anchor bolt pocket and grout key clearance

3. REBAR DETAILING & PLACEMENT MANDATES:

- All deformed reinforcing bars shall conform to ASTM A615 Grade 60 with minimum yield strength Fy = 414 MPa.
- Concrete spacers and chairs (precast concrete blocks with tied wires) shall be placed at 1,000mm maximum spacing.
- Rebars shall not be bent or straightened in a manner that will injure or weaken the material. No field heating.
- All lap splices must be staggered by at least 600mm. No splices allowed in maximum flexural moment zones.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

SHEET TITLE: FOUNDATION REINFORCEMENT & REBAR SECTIONAL ARRAN

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_ST_003	FOR TENDER / TYPICAL	R0 / 26-OCT-26	09 OF 30

BILL OF QUANTITIES (BOQ) & BAR BENDING SCHEDULE (BBS)

ITEMIZED REINFORCING CUTTING SCHEDULE • UNIT WEIGHTS • CONCRETE VOLUMES • EMBEDMENTS

COMPLETE BAR BENDING SCHEDULE (BBS) - FOUNDATION MAT & PEDESTAL

MARK	LOCATION / DESCRIPTION	BAR DIA	SHAPE CODE	CUTTING LENGTH	NO. OF BARS	TOTAL LENGTH	UNIT WT	TOTAL WT (kg)
F1-A	Mat Bottom Grid (X-Dir)	20 mm	Code 21 (U-Hooked)	5,350 mm	34 pcs	181.9 m	2.47 kg/m	449.3 kg
F1-B	Mat Bottom Grid (Y-Dir)	20 mm	Code 21 (U-Hooked)	5,350 mm	34 pcs	181.9 m	2.47 kg/m	449.3 kg
F2-A	Mat Top Grid (X-Dir)	16 mm	Code 21 (U-Hooked)	5,300 mm	34 pcs	180.2 m	1.58 kg/m	284.7 kg
F2-B	Mat Top Grid (Y-Dir)	16 mm	Code 21 (U-Hooked)	5,300 mm	34 pcs	180.2 m	1.58 kg/m	284.7 kg
P1	Pedestal Chimney Verticals	25 mm	Code 11 (L-Hooked)	2,450 mm	20 pcs	49.0 m	3.85 kg/m	188.7 kg
P2-A	Pedestal Ties (Upper Zone)	12 mm	Code 51 (Closed Rect)	6,900 mm	6 pcs	41.4 m	0.89 kg/m	36.8 kg
P2-B	Pedestal Ties (Lower Zone)	12 mm	Code 51 (Closed Rect)	6,900 mm	8 pcs	55.2 m	0.89 kg/m	49.1 kg
CH1	Rebar Chairs & Continuous Spacers	16 mm	Code 99 (Chairs)	1,800 mm	25 pcs	45.0 m	1.58 kg/m	71.1 kg
TOTAL	TOTAL REINFORCING STEEL (GRADE 60)	--	--	--	225 pcs	864.8 m	--	1,813.7 kg

CONCRETE & CIVIL MATERIAL TAKE-OFF SUMMARY

ITEM DESCRIPTION	SPECIFICATION / STRENGTH	NEAT VOLUME	WASTAGE	TOTAL ORDER
Structural Concrete (Mat)	f'c >= 28 MPa (4,000 psi) @ 28d	20.00 m3	5%	21.00 m3
Structural Concrete (Pedestal)	f'c >= 28 MPa (4,000 psi) @ 28d	3.89 m3	5%	4.10 m3
Blinding Lean Concrete	f'c >= 15 MPa (2,000 psi) @ 28d	1.35 m3	10%	1.50 m3
Crushed Stone Bedding (G-1)	Clean crushed stone 20-40mm	6.40 m3	10%	7.04 m3
Non-Shrink Structural Grout	f'g >= 50 MPa (Pre-packaged)	0.08 m3	15%	12 Bags (25kg)

STEEL EMBEDMENTS & HARDWARE SCHEDULE

HARDWARE COMPONENT	MATERIAL SPECIFICATION	SIZE / RATING	QUANTITY
Anchor Bolts Assembly	ASTM F1554 Grade 105 High-Strength	M36 x 1,800mm (all-thread)	12 Nos.
Baseplate Leveling Nuts	ASTM A563 Grade DH Heavy Hex	M36 Heavy Hex Zinc Plated	12 Nos.
Baseplate Top Double Nuts	ASTM A563 Grade DH Heavy Hex	M36 Double Nuts + Washers	24 Nos.
Anchor Bottom Washers	ASTM A36 Steel Washer Plate	150mm x 150mm x 20mm thk	12 Nos.
Cadweld Grounding Terminals	Bare Copper Stranded Conductor	50 mm2 conductor to rebar	4 Points

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)

SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

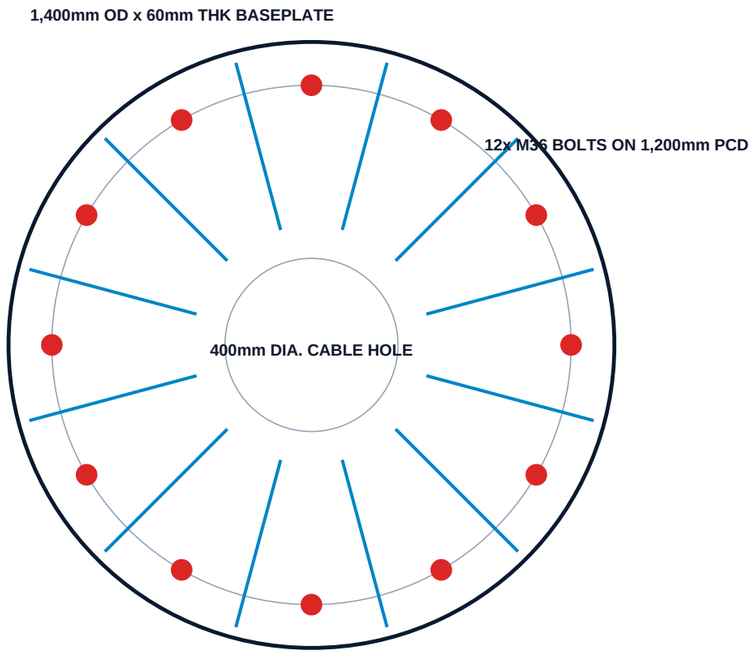
SHEET TITLE: FOUNDATION REBAR BENDING SCHEDULE & MATERIAL TAK

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_ST_004	FOR TENDER / TYPICAL	R0 / 26-OCT-26	10 OF 30

MONOPOLE FLANGE BASEPLATE & M36 ANCHOR BOLT DETAILS

1,400MM OD x 60MM THK PLATE • 12x M36 GRADE 105 BOLTS • 12x 25MM GUSSETS • GROUT BED

BASEPLATE TEMPLATE & ANCHOR BOLT PLAN (SCALE 1:20)



ANCHOR BOLT & BASEPLATE ENGINEERING SPECIFICATIONS

1. BASEPLATE GEOMETRICAL & MECHANICAL PARAMETERS:

PARAMETER	SPECIFIED VALUE	DESIGN CRITERIA / CODE
Baseplate Outside Diameter	1,400 mm (+/- 3mm)	Matches pedestal clearance envelope
Baseplate Plate Thickness	60.0 mm Solid Plate	ASTM A572 Grade 50 (Fy = 345 MPa)
Bolt Pitch Circle Dia (PCD)	1,200 mm (+/- 1.5mm)	Strict template jig fabrication required
Internal Cable Feed Port	400 mm Diameter	Smooth ground edges for cable protection
Radial Stiffener Gussets	12 Nos. (25mm thk x 250mm H)	Welded with full penetration bevel welds
Grout Bed Thickness	50.0 mm (+/- 5mm)	Non-shrink high-strength grout (f'g >= 50 MPa)

2. M36 ANCHOR BOLT CAPACITY & UTILIZATION (ASTM F1554 GR 105):

FORCE / CAPACITY CHECK	NOMINAL VALUE	APPLIED ULTIMATE	UTILIZATION	STATUS
Max Tensile Force per Bolt	Phi*Pn = 620 kN	Pu = 445 kN	0.72	PASS (Safe)
Max Shear Force per Bolt	Phi*Vn = 285 kN	Vu = 24.2 kN	0.08	PASS (Safe)
Combined Tension + Shear	Interaction <= 1.0	(P/Pn)^2 + (V/Vn)^2	0.53	PASS (Safe)
Embedment Depth in Concrete	1,450 mm Required	1,450 mm Provided	1.00	PASS (Developed)
Concrete Breakout Capacity	Phi*Ncb = 820 kN	Nu = 445 kN	0.54	PASS (Safe)

3. ANCHOR BOLT INSTALLATION MANDATES:

- A rigid steel template jig (top and bottom) must hold all 12 bolts securely in place during concrete pouring.
- Bolt threads must be fully protected from slurry and physical damage during concrete casting and curing.
- Leveling nuts below the baseplate shall be adjusted using optical surveyor level before monopole erection.
- Top nuts shall be tightened to 1/3 turn beyond snug-tight condition using calibrated torque wrench.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

SHEET TITLE: BASEPLATE TEMPLATE & HOLD-DOWN ANCHOR BOLT ASSEM

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_ST_005	FOR TENDER / TYPICAL	R0 / 26-OCT-26	11 OF 30

GENERAL STRUCTURAL CONCRETE & CIVIL CONSTRUCTION SPECIFICATIONS

NSCP 2015 / ACI 318-19 QUALITY CONTROL • MIX DESIGN • PLACEMENT • CURING • TESTING

TECHNICAL SPECIFICATIONS FOR REINFORCED CONCRETE WORKS

1. MATERIALS SPECIFICATIONS & QUALITY ASSURANCE:

- PORTLAND CEMENT: ASTM C150 Type I normal Portland cement. Shall be fresh, dry, and free of lumps.
- FINE AGGREGATE: Clean, hard, natural silica sand conforming to ASTM C33. Free of organic impurities.
- COARSE AGGREGATE: Crushed basalt or andesite gravel conforming to ASTM C33, max nominal size 25mm (1 inch).
- MIXING WATER: Clean, potable water, free from injurious amounts of oils, acids, alkalis, salts, and organics.
- REINFORCING BARS: ASTM A615 Grade 60 deformed steel bars (Fy >= 414 MPa). Mill test certificates required.
- ADMIXTURES: ASTM C494 Type A (Water-reducing) or Type F (High-range). No calcium chloride permitted.
- NON-SHRINK GROUT: Pre-blended, non-metallic, cementitious grout meeting ASTM C1107 (f'g >= 50 MPa).

2. CONCRETE MIX DESIGN & COMPRESSIVE STRENGTH REQUIREMENTS:

- FOUNDATION MAT & PEDESTAL: Class 'A' concrete with minimum f'c = 28 MPa (4,000 psi) at 28 days.
- SLUMP TOLERANCE: 100mm +/- 25mm without superplasticizer; 150mm +/- 25mm with high-range water reducer.
- MAXIMUM WATER-CEMENT RATIO: Maximum w/c ratio shall not exceed 0.45 by weight for durability.
- AIR CONTENT: 3.5% to 5.0% entrapped air content for standard mass concrete foundations.
- BLINDING LEAN CONCRETE: Subgrade seal concrete with minimum f'c = 15 MPa (2,000 psi) at 28 days.

3. EXCAVATION, SHORING & SUBGRADE PREPARATION:

- EXCAVATION: Trim all excavation bottoms to neat lines. Remove all loose, disturbed, or organic material.
- SHORING: Provide adequate temporary timber or steel trench shoring if excavation exceeds 1.5m depth.
- SUBGRADE COMPACTION: Compact founding subgrade to minimum 95% Modified Proctor Density (ASTM D1557).
- DEWATERING: Sump pits and submersible pumps must keep excavation completely dry during concrete casting.

4. CONCRETE PLACEMENT & CONSOLIDATION:

- FREE FALL LIMIT: Concrete shall not be dropped freely from a height exceeding 1.50 meters.
- CONSOLIDATION: Internal immersion vibrators (minimum 2.5-inch head, >= 8,000 vpm) shall be used continuously.
- INSERTION SPACING: Vibrators shall penetrate vertically into previous layer at 450mm centers for 10-15 seconds.
- COLD JOINTS: Continuous pouring required for mat footing. No construction joints permitted within mat block.
- TEMPERATURE CONTROL: Maximum concrete temperature during discharge shall not exceed 32 degrees Celsius.

5. CURING & FORMWORK REMOVAL:

- MOIST CURING: Keep all exposed concrete continuously damp for at least 7 consecutive days using wet burlap.
- CURING COMPOUNDS: Liquid membrane-forming curing compounds conforming to ASTM C309 may be applied.
- SIDE FORMS REMOVAL: Foundation side formwork shall not be stripped earlier than 24 hours after pouring.
- BACKFILLING TIMELINE: Structural backfill over footing mat shall not commence until concrete reaches 70% f'c.

6. FIELD SAMPLING & COMPRESSION TESTING PROTOCOL:

- SAMPLING RATE: One set of four (4) standard 150mm x 300mm cylinders for every 20 m3 of concrete cast.
- TESTING AGES: One cylinder tested at 7 days; two cylinders tested at 28 days; one cylinder kept as reserve.
- ACCEPTANCE CRITERIA: Average of any 3 consecutive 28-day tests must equal or exceed specified f'c (28 MPa).
- NON-COMPLIANCE: Concrete failing strength tests shall require core testing (ASTM C42) at contractor's expense.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC.	TELTECH 5G FWA
PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50) SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa	
LEAD TECHNICAL ARCHITECT / REP: OLIVER TUNGOL	PROFESSIONAL STRUCTURAL ENGINEER: JOMAR DOGILLO, PE
SHEET TITLE: GENERAL CONCRETE SPECIFICATIONS & CIVIL CONSTRUC	
DRAWING NO: TT_ST_006	RELEASE STATUS: FOR TENDER / TYPICAL
REV / DATE: R0 / 26-OCT-26	SHEET NO: 12 OF 30

STRUCTURAL STEEL FABRICATION, WELDING & GALVANIZING SPECIFICATIONS

AISC 360-16 • AWS D1.1:2020 • ASTM A123 / ISO 1461 • ASTM A572 GR 50 • NDT QUALITY CONTROL

TECHNICAL SPECIFICATIONS FOR STRUCTURAL STEELWORKS

1. STRUCTURAL STEEL MATERIAL STANDARDS:

- MONOPOLE SHAFT PLATES: ASTM A572 Grade 50 (Fy = 345 MPa, Fu = 450 MPa). Certified mill test reports mandatory.
- BASEPLATE & FLANGES: ASTM A572 Grade 50 solid structural steel plate, Ultrasonic tested for laminations (ASTM A435).
- GUSSET STIFFENERS: ASTM A572 Grade 50 structural steel plate, beveled for full-penetration welding.
- MISCELLANEOUS BRACKETS: ASTM A36 structural steel for secondary mounts, ladder rungs, and gantry supports.
- STRUCTURAL BOLTS & NUTS: High-strength structural bolts ASTM A325 or Grade 8.8, hot-dip galvanized (ASTM A153).
- ANCHOR RODS: ASTM F1554 Grade 105 high-strength heat-treated alloy steel all-thread rods (Fy = 724 MPa, Fu = 862 MPa).

2. WELDING PROCEDURES & WELDER QUALIFICATIONS (AWS D1.1):

- WELDING CODE: All welding shall strictly conform to AWS D1.1/D1.1M Structural Welding Code - Steel.
- WELDING PROCESSES: Submerged Arc Welding (SAW) for longitudinal seam welds; SMAW/GMAW for attachments.
- ELECTRODES: Low-hydrogen electrodes conforming to AWS A5.1 / A5.5 Class E7018. Oven storage at 120 deg C required.
- LONGITUDINAL SEAM WELDS: Minimum 100% full-penetration complete joint penetration (CJP) in slip joint lap zones.
- PREHEAT REQUIREMENTS: Base metal must be preheated to minimum 65 deg C (150 deg F) for plates > 25mm thickness.
- WELDER CERTIFICATION: All welders and welding operators shall be certified in accordance with AWS D1.1 Chapter 4.

3. NON-DESTRUCTIVE TESTING (NDT) & INSPECTION:

- VISUAL INSPECTION: 100% visual inspection of all welds in accordance with AWS D1.1 Table 6.1.
- ULTRASONIC TESTING (UT): 100% UT testing on all longitudinal CJP welds within male and female slip joint zones.
- MAGNETIC PARTICLE (MT): 20% MT testing on baseplate-to-shaft CJP welds and gusset stiffener fillet welds.
- DEFECT REPAIR: All rejected welds shall be gouged, repaired, and re-tested using identical NDT procedures.

4. HOT-DIP GALVANIZING SPECIFICATIONS (ASTM A123):

- ZINC COATING STANDARD: Hot-dip galvanized in accordance with ASTM A123/A123M after all fabrication is complete.
- MINIMUM COATING THICKNESS: Average coating thickness >= 85 microns (610 g/m2) for structural steel plates > 6mm.
- PRE-TREATMENT: Acid pickling in dilute sulfuric or hydrochloric acid, thoroughly fluxed before zinc kettle immersion.
- VENTILATION HOLES: Internal drain and vent holes must be provided at all closed tube ends to prevent kettle explosion.
- GALVANIZING REPAIR: Scratches and damaged zinc areas shall be repaired using zinc-rich paint (ASTM A780, >= 92% zinc).

5. SHOP ASSEMBLY, MATCH-MARKING & SLIP JOINTS:

- TRIAL SHOP ASSEMBLY: All five (5) sections shall be pre-assembled in horizontal alignment at fabrication facility.
- MATCH-MARKING: Permanent steel die-stamped orientation marks must be applied across each slip-joint interface.
- CLAPPING DEPTH: Slip joint insertion length shall be scribed permanently on male section prior to galvanizing.
- JACKING ATTACHMENTS: Welded jacking lugs or jacking clamp collars must be provided for hydraulic erection pull-up.

6. PACKING, HANDLING & TRANSPORTATION:

- PROTECTIVE CRIBBING: Monopole sections must be supported on padded timber cribbing during storage and transit.
- NYLON SLINGS: Only nylon or synthetic webbing slings shall be used for rigging to prevent galvanizing damage.
- STORAGE CLEARANCE: Galvanized components stored at job site must be elevated off damp soil to avoid wet-storage stain.

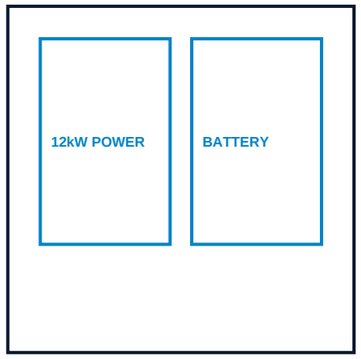
TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC.	TELTECH 5G FWA
PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50) SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa	
LEAD TECHNICAL ARCHITECT / REP: OLIVER TUNGOL	PROFESSIONAL STRUCTURAL ENGINEER: JOMAR DOGILLO, PE
SHEET TITLE: STRUCTURAL STEEL FABRICATION, WELDING & GALVANIZ	
DRAWING NO: TT_ST_007	RELEASE STATUS: FOR TENDER / TYPICAL
REV / DATE: R0 / 26-OCT-26	SHEET NO: 13 OF 30

OUTDOOR EQUIPMENT CONCRETE PAD PLINTH DETAILS (2.0M x 2.0M)

ELEVATED +300MM AGL • REINFORCING MAT • M16 ANCHORS • 12KW POWER PLANT & BATTERIES

EQUIPMENT PAD STRUCTURAL PLAN & SECTION (SCALE 1:25)

PLINTH PLAN VIEW (2.0M x 2.0M PAD):



PLINTH CROSS SECTION VIEW:



EQUIPMENT PLINTH SPECIFICATIONS & LOADS

1. PLINTH STRUCTURAL PARAMETERS:

PARAMETER	SPECIFICATION	REMARKS / DETAILS
Footprint Dimensions	2.00m x 2.00m (+/- 10mm)	4.00 m2 surface area
Finished Thickness	300 mm Solid Concrete	+300mm elevation above compound grade
Concrete Strength	f'c >= 28 MPa @ 28 Days	Class 'A' Portland cement concrete
Reinforcement Mat	12mm dia. @ 150mm c/c	Top & bottom mats (ASTM A615 Gr 60)
Subgrade Base	100mm Clean Gravel (G-1)	Compacted on approved subgrade
Edge Chamfer	25mm x 25mm 45 deg	All exposed top perimeters chamfered
Cabinet Hold-down Anchors	8x M16 316-SS Anchors	Embedment depth >= 120mm

2. SUPPORTED OUTDOOR EQUIPMENT LOADS:

EQUIPMENT ENCLOSURE	QTY	DIMENSIONS (mm)	WEIGHT (OP)	HEAT DISSIPATION
12kW Telecom Power Plant	1 Unit	650 W x 650 D x 1600 H	180 kg	850 Watts
200Ah/48V LiFePO4 Battery Bank	2 Banks	650 W x 650 D x 1200 H	220 kg/ea	Passive Cooling
AC Power Distribution Box (ACPDB)	1 Unit	400 W x 250 D x 600 H	35 kg	Negligible
Main DC Distribution Bus (MDP)	1 Unit	Integrated in Cabinet	25 kg	150 Watts
TOTAL EQUIPMENT LOAD	--	--	680 kg	1,000 Watts

3. PLINTH CIVIL DIRECTIVES:

- Plinth elevation (+300mm) provides vital flood mitigation against overland sheet flow and ponding.
- PVC electrical conduits (50mm dia.) must be pre-cast into the pad with sweeping 90-degree elbows.
- A 50mm2 bare copper ground conductor must be embedded into the pad rebar cage for equipotential bonding.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — **TELTECH 5G FWA**

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

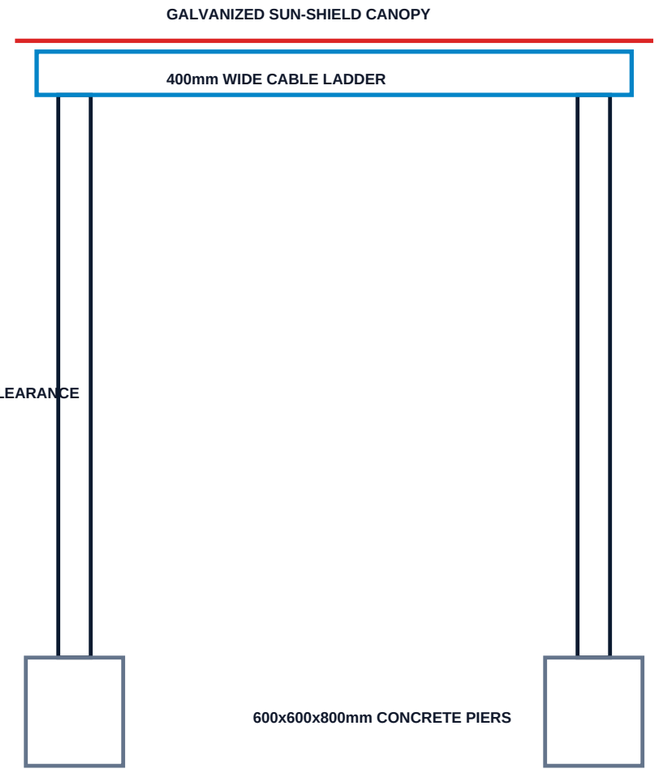
SHEET TITLE: OUTDOOR EQUIPMENT PAD PLINTH DETAILS (2.0M x 2.0

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_ST_008	FOR TENDER / TYPICAL	R0 / 26-OCT-26	14 OF 30

ELEVATED CABLE GANTRY & CABLE TRAY STRUCTURAL DETAILS

3.5M SPAN • +2.5M CLEARANCE • 100x100 RHS POSTS • 400MM LADDER • SUN-SHIELD CANOPY

ELEVATED CABLE GANTRY ELEVATION & PROFILE (SCALE 1:20)



CABLE GANTRY STRUCTURAL SPECIFICATIONS

1. GANTRY STRUCTURAL MEMBERS:

STRUCTURAL MEMBER	SECTION / SIZE	MATERIAL / COATING
Vertical Support Columns	100mm x 100mm x 4.5mm RHS	ASTM A500 Gr. B / HDG >= 85um
Cross-Beam Support Traverses	75mm x 50mm x 4.0mm Channel	ASTM A36 / HDG >= 85um
Cable Ladder Tray	400mm W x 100mm H Rung Tray	Pre-galvanized / HDG Steel
Protective Sun Canopy	1.2mm Corrugated Aluminum	Snap-on weather cover
Column Baseplates	250mm x 250mm x 12mm Plate	ASTM A36 with 4-M16 anchor bolts
Concrete Column Footings	600mm x 600mm x 800mm Deep	f'c >= 28 MPa with 4-12mm rebar

2. CABLE CAPACITY & ROUTING SCHEDULE:

CABLE CATEGORY	CABLE TYPE / OD	QUANTITY	TRAY CAPACITY
Band n50 RF Jumpers	1/2-inch Superflexible (16mm OD)	6 Runs	15% fill ratio
RRU DC Power Feeds	2x16 mm2 Shielded DC (14mm OD)	3 Runs	8% fill ratio
Backhaul Optical Fiber	48-Core Armored (12mm OD)	2 Runs	4% fill ratio
Starlink Cat6A STP	Outdoor UV-Rated (8mm OD)	1 Run	2% fill ratio
Grounding Equipotential	25mm x 3mm Tinned Copper Tape	1 Continuous	Dedicated Clamp
TOTAL TRAY UTILIZATION	400mm Wide Perforated Ladder	--	29% (Expansion Ready)

3. GANTRY INSTALLATION MANDATES:

- All RF and optical cables must be secured to rungs using UV-resistant stainless steel cable cleats.
- Cables must transition into tower and cabinets with minimum bending radius >= 15 times cable outer diameter.
- Continuous equipotential ground tape (25x3mm) must bond both gantry ends directly to the MGB.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — **TELTECH 5G FWA**

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

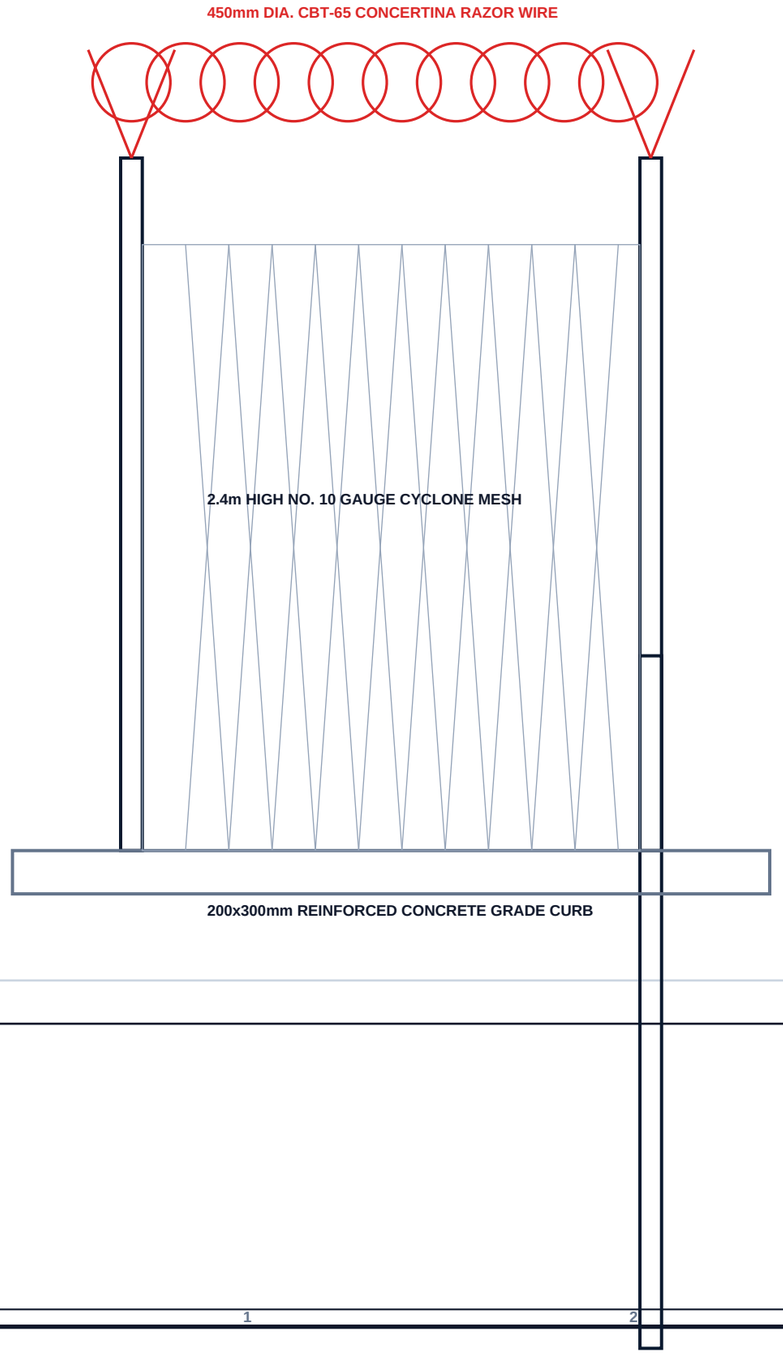
SHEET TITLE: ELEVATED CABLE GANTRY & CABLE LADDER STRUCTURAL

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_ST_010	FOR TENDER / TYPICAL	R0 / 26-OCT-26	15 OF 30

COMPOUND PERIMETER SECURITY FENCING MASTER DETAILS

2.4M CYCLONE MESH (NO. 10 GA) • 0.5M Y-ARMS • 450MM CONCERTINA RAZOR WIRE • 2.9M TOTAL HEIGHT

FENCE PANEL ELEVATION & RAZOR WIRE DETAILS (SCALE 1:30)



SECURITY FENCING TECHNICAL SPECIFICATIONS

1. FENCE STRUCTURAL SPECIFICATIONS:

COMPONENT	DIMENSIONS / SCHEDULE	SPECIFICATION
Total Fence Height	2.90 Meters Total	2.4m fabric + 0.5m concertina Y-arm
Chain Link Fabric	No. 10 Gauge (3.2mm core)	50mm x 50mm diamond mesh (HDG)
Fence Line Posts	50mm dia. (60.3mm OD)	Schedule 40 Galvanized Pipe @ 2.0m c/c
Top, Mid & Bot Rails	40mm dia. (48.3mm OD)	Schedule 40 Galvanized Pipe continuous
Concertina Extension Arms	45 deg Y-Crank Extension	Pressed steel hot-dip galvanized
Concertina Razor Wire	450mm dia. CBT-65 Profile	AISI 430 Stainless steel high-tensile core
Barbed Wire Strands	3 Strands (12-4-5-4)	Galvanized high-tensile steel wire

2. PERIMETER BILL OF QUANTITIES (8.0M x 8.0M):

BILL ITEM	QUANTITY	UNIT	REMARKS
Chain-Link Mesh Fabric	76.8	Sq. Meters	32.0m perimeter x 2.4m height
Schedule 40 Line Posts	16	Pieces	60.3mm OD x 3.2m total length
Schedule 40 Rails	96.0	Linear Meters	Top, middle, and bottom rails
Concertina Razor Wire	32.0	Linear Meters	450mm dia. CBT-65 continuous coil
Galvanized Barbed Wire	96.0	Linear Meters	3 continuous strands along Y-arms
Concrete Grade Curb	1.92	Cubic Meters	200mm W x 300mm D continuous curb

3. FENCING INSTALLATION DIRECTIVES:

- Line posts shall be embedded at least 800mm into reinforced concrete footings cast integral with grade curb.
- Chain link mesh shall be tightly stretched with mechanical come-along winch and secured with tension bars.
- The entire perimeter fence must be bonded to the site earthing ring at each of the 4 corner posts.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

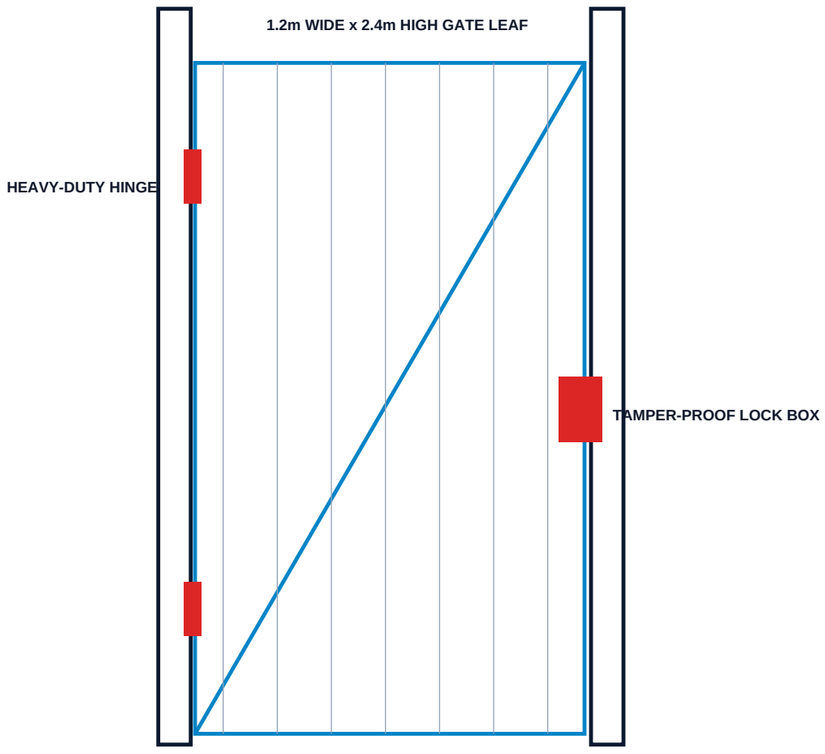
SHEET TITLE: COMPOUND SECURITY FENCING DETAILS (CYCLONE MESH

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_ST_011	FOR TENDER / TYPICAL	R0 / 26-OCT-26	16 OF 30

COMPOUND ENTRANCE PEDESTRIAN / SERVICE SWING GATE DETAILS

1.2M WIDE x 2.4M HIGH • HEAVY-DUTY BARREL HINGES • SLIDE BOLT • SHROUDED PADLOCK BOX

GATE ELEVATION & LOCKING DETAILS (SCALE 1:25)



GATE SPECIFICATIONS & HARDWARE SCHEDULE

1. GATE STRUCTURAL SPECIFICATIONS:

GATE COMPONENT	MEMBER SIZE / SPEC	MATERIAL / GRADE
Gate Hinge & Latch Posts	75mm dia. (88.9mm OD)	Schedule 40 Pipe / HDG >= 85um
Gate Leaf Outer Frame	50mm dia. (60.3mm OD)	Schedule 40 Pipe / HDG >= 85um
Internal Diagonal Brace	40mm dia. (48.3mm OD)	Schedule 40 Pipe full penetration welded
Chain Link Mesh Infill	No. 10 Gauge Cyclone Mesh	50mm x 50mm diamond mesh
Heavy-Duty Barrel Hinges	25mm dia. Pin with Grease Zerk	Weld-on steel hinges (2 Nos.)
Tamper-Proof Lock Box	6mm Solid Steel Shrouded Box	Protects padlock hasp from bolt cutters
Drop-Rod Cane Bolt	20mm dia. Solid Steel Rod	Locks gate in open or closed position

2. GATE CLEARANCE & FOUNDATION PARAMETERS:

CLEARANCE PARAMETER	SPECIFIED VALUE	DESIGN CRITERIA
Clear Ingress Width	1,200 mm Clear	Accommodates personnel & equipment replacement
Underside Clearance	50 mm above Grade Curb	Prevents dragging while denying animal intrusion
Gate Post Concrete Pier	600mm x 600mm x 1,000mm	f'c >= 28 MPa with 4-16mm rebar cage
Grounding Bonding Jumper	50 mm2 Flexible Copper Braid	Bonds gate leaf across hinges to fence post

3. GATE SECURITY DIRECTIVES:

- The gate shall swing inward into the compound to avoid obstruction of the external access corridor.
- Padlocks must be carrier-grade heavy brass with boron-carbide shackle, master-keyed to TelTech regional SAQ.
- Flexible grounding braid (50mm2) must bridge the hinge gap to ensure zero static potential on the gate leaf.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

SHEET TITLE: COMPOUND ENTRANCE SWING GATE & SECURITY LOCKING

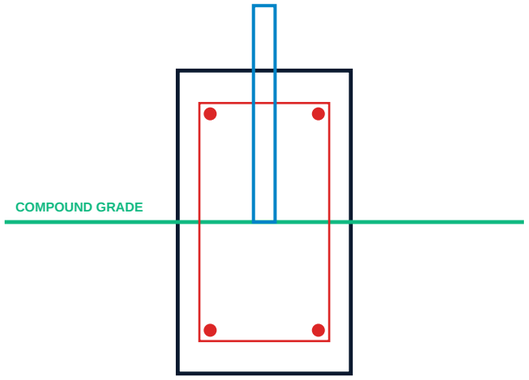
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TT_ST_012	FOR TENDER / TYPICAL	R0 / 26-OCT-26	17 OF 30

CONTINUOUS CONCRETE GRADE CURB & CORNER BRACING DETAILS

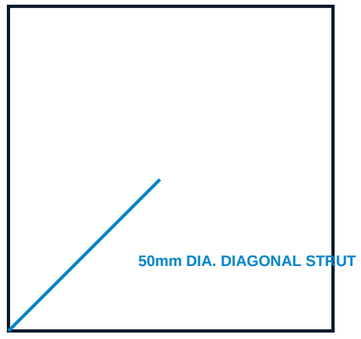
200MM x 300MM CURB • 4-12MM REBAR • 45 DEG CORNER BRACE STRUTS • SOIL EROSION CONTROL

GRADE CURB & CORNER STRUT DETAILS (SCALE 1:20)

CONTINUOUS GRADE CURB CROSS SECTION:



CORNER POST 45 DEGREE BRACE STRUT PLAN:



GRADE CURB & BRACING TECHNICAL SPECIFICATIONS

1. REINFORCED GRADE CURB SPECIFICATIONS:

COMPONENT	DIMENSION / SPECIFICATION	MANDATE
Curb Cross-Section	200mm Wide x 300mm Deep	Continuous along all 32m of perimeter
Embedment Depth	150mm below grade, 150mm above	Prevents burrowing & soil washout
Concrete Specification	f'c >= 28 MPa @ 28 Days	Class 'A' structural concrete
Longitudinal Rebar	4 Nos. 12mm dia. Rebars	ASTM A615 Grade 60 (Fy = 414 MPa)
Lateral Ties / Stirrups	10mm dia. @ 200mm c/c	Closed rectangular ties with 135 deg hooks
Drainage Weep Holes	50mm dia. PVC @ 2.0m c/c	Permits internal storm drainage egress

2. CORNER & TERMINAL BRACING SPECIFICATIONS:

BRACING MEMBER	SIZE / SCHEDULE	LOCATION / QUANTITY
Corner Post Struts	50mm dia. Schedule 40 Pipe	2 Struts at each of 4 corners (8 total)
Gate Post Terminal Struts	50mm dia. Schedule 40 Pipe	1 Strut behind each gate post (2 total)
Brace Angle & Slope	45.0 Degrees from Horizontal	Anchored into curb / independent pier
Brace Clamping Bands	Heavy Galvanized Steel Bands	Bolted with 12mm HDG machine bolts

3. CURB & STRUT MANDATES:

- Continuous curb casting prevents fence undermining from storm run-off and rural burrowing fauna.
- Weep holes must have stainless steel mesh screens on the interior face to prevent pest ingress.
- Expansion joints (15mm asphalt impregnated fiber board) shall be installed every 8.0 meters.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

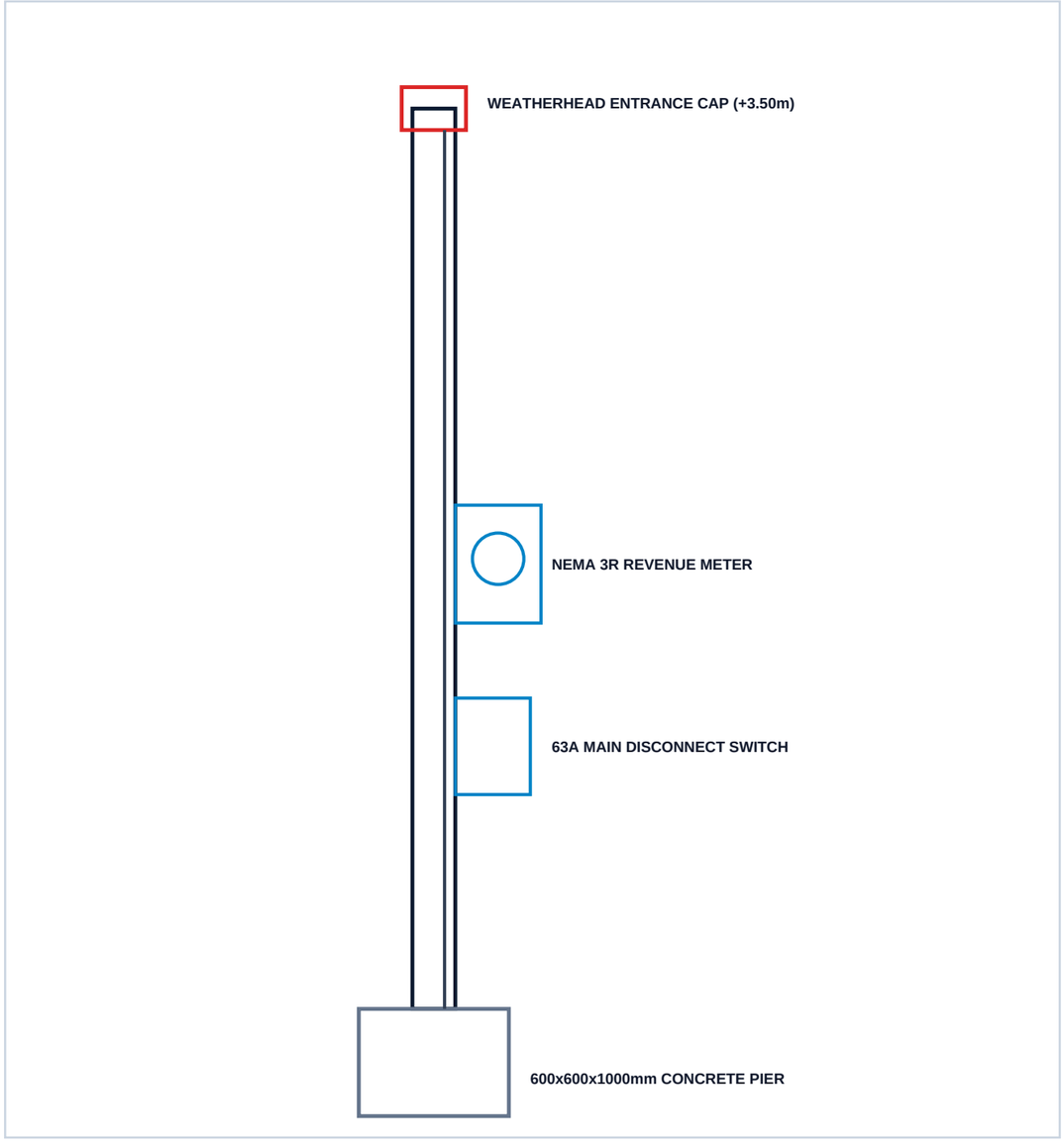
SHEET TITLE: FENCE CORNER POST STIFFENERS & REINFORCED GRADE

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_ST_013	FOR TENDER / TYPICAL	R0 / 26-OCT-26	18 OF 30

230V UTILITY SERVICE ENTRANCE & POWER METERING PEDESTAL

+3.5M AGL 150x150 SHS POST • REVENUE METER • 63A MAIN DISCONNECT • WEATHERHEAD

SERVICE ENTRANCE PEDESTAL ELEVATION (SCALE 1:25)



UTILITY SERVICE ENTRANCE SPECIFICATIONS

1. SERVICE ENTRANCE PEDESTAL SPECIFICATIONS:

EQUIPMENT / COMPONENT	RATING / SPECIFICATION	STANDARDS
Service Entrance Column	150mm x 150mm x 5.0mm SHS	ASTM A500 Gr. B / HDG >= 85um
Weatherhead Service Cap	50mm dia. Aluminum Cap	NEMA / UL listed outdoor type
Service Riser Conduit	50mm dia. Rigid Steel (RSC)	ANSI C80.1 Hot-Dip Galvanized
Revenue Meter Base	100A, 230V 1-Ph 60Hz Socket	Electric Co-op / DU Approved (IP65)
Main Service Disconnect	63A 2-Pole MCB in NEMA 3R Box	IEC 60947-2 (10kA IC rating)
Concrete Pier Foundation	600mm x 600mm x 1,000mm Deep	f'c >= 28 MPa with 4-M20 anchor bolts

2. UTILITY SERVICE INTERCONNECTION PARAMETERS:

PARAMETER	VALUE	REMARKS / DETAILS
Nominal Supply Voltage	230 Volts AC (+/- 10%)	Single-Phase, 2-Wire + Ground, 60 Hz
Service Drop Clearance	Minimum 5.50m over road	DPWH / PEC minimum overhead clearance
Connected Load	12.0 kW (Full Rectifier Load)	Nominal operating load: 4.8 kW
Overcurrent Protection	63A 2-Pole Main Breaker	Thermal-magnetic trip coordination

3. ELECTRICAL SAFETY MANDATES:

- The neutral conductor must remain completely isolated from the grounding conductor at the meter pedestal.
- Service entrance post must be bonded to the perimeter earthing ring using a 50mm2 bare copper lead.
- A padlock provision must be provided on the meter door for distribution utility (DU) sealing.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

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LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

SHEET TITLE: SERVICE ENTRANCE PEDESTAL & 230V UTILITY METERIN

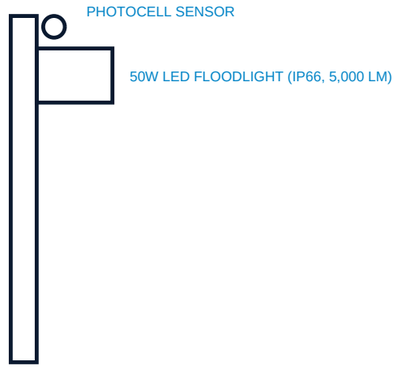
DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_ST_014	FOR TENDER / TYPICAL	R0 / 26-OCT-26	19 OF 30

PERIMETER SECURITY LIGHTING & SAFETY FIRE PROTECTION DETAILS

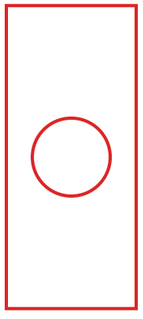
50W LED FLOODLIGHTS • DUSK-TO-DAWN PHOTOCELL • NEMA 4 FIRE ENCLOSURE • 10-LB CO2

PERIMETER LIGHTING & FIRE ENCLOSURE ELEVATION (SCALE 1:20)

50W LED SECURITY FLOODLIGHT MOUNTING (+2.8M AGL):



WEATHERPROOF FIRE EXTINGUISHER ENCLOSURE:



10-LB (4.5 KG) CO2 FIRE EXTINGUISHER (CLASS B & C)

SAFETY & PERIMETER LIGHTING SPECIFICATIONS

1. SECURITY LIGHTING FIXTURE SPECIFICATIONS:

PARAMETER	SPECIFICATION	OPERATIONAL DETAILS
Luminaire Type	50W High-Efficiency LED Floodlight	5,000 Lumens luminous output
Ingress Protection	IP66 Weatherproof / Dust-tight	Die-cast aluminum housing, tempered glass
Color Temperature	5,700K Daylight White	CRI >= 80 for clear CCTV surveillance
Automatic Control	Dusk-to-Dawn Photocell Sensor	Automatic switching with manual bypass switch
Mounting Elevation	+2.80 Meters AGL	Clamped to diagonal opposite fence corner posts
Electrical Supply	230V AC 60Hz from ACPDB	Protected by dedicated 15A 1-Pole MCB

2. SITE FIRE PROTECTION SPECIFICATIONS:

EQUIPMENT	RATING / CAPACITY	MANDATE / STANDARD
Fire Extinguisher	10-lb (4.5 kg) Carbon Dioxide (CO2)	Class B & C electrical fire certified
Enclosure Cabinet	NEMA 4 Heavy-Gauge Steel (Red)	Break-glass front door or padlock provision
Inspection Tag	Annual BFP Verification Tag	Bureau of Fire Protection compliance
Mounting Location	Service Post or Entrance Gate	Immediately accessible upon entry

3. SAFETY DIRECTIVES:

- Light fixtures must be oriented downward at 30 degrees to minimize light pollution into adjacent properties.
- CO2 fire extinguishers are mandatory to prevent dry chemical residue damage to sensitive telecom electronics.
- Monthly visual inspection of pressure gauges and annual recharging must be logged in site maintenance logbook.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

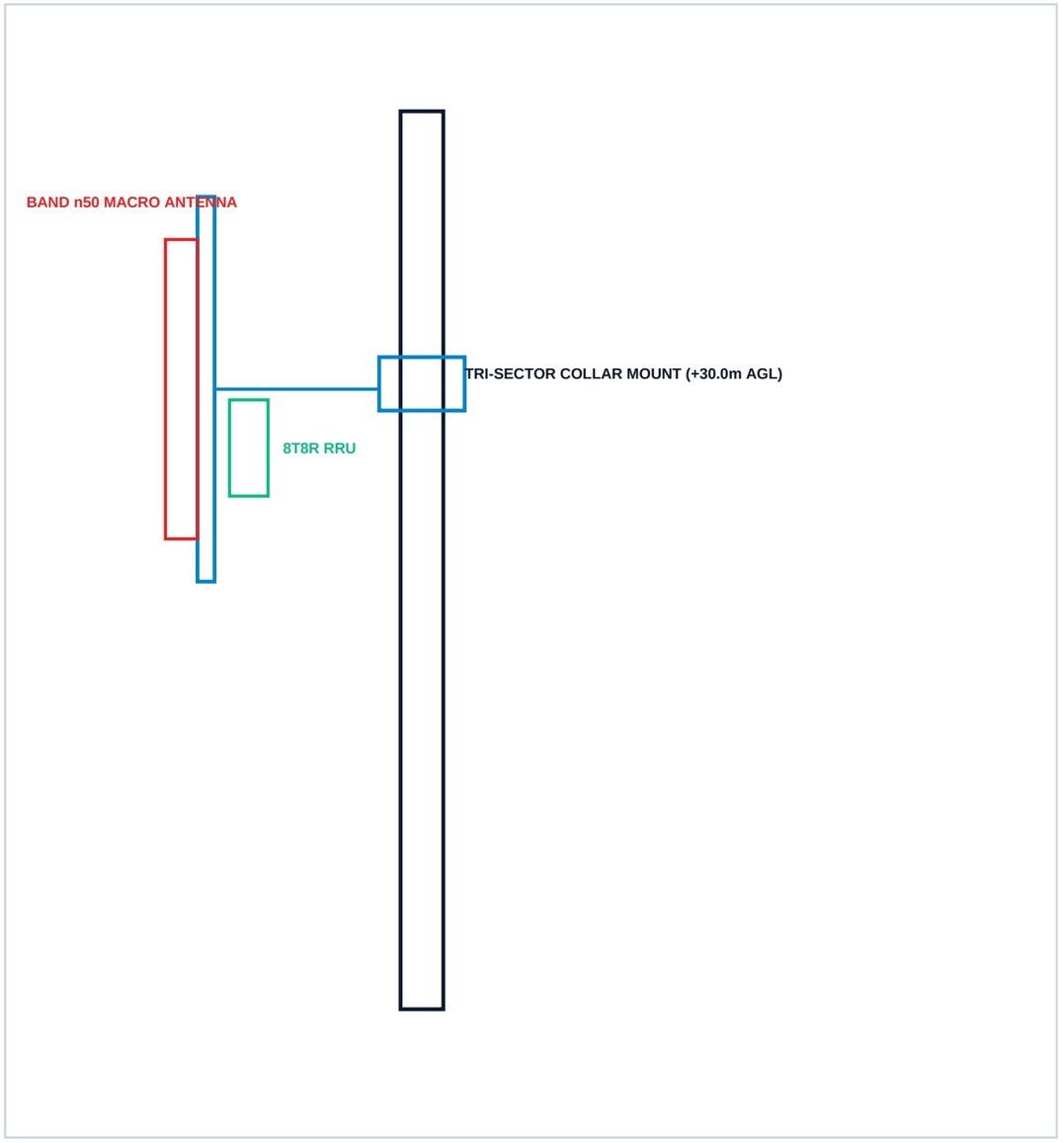
SHEET TITLE: PERIMETER SECURITY LIGHTING & FIRE EXTINGUISHER

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_ST_015	FOR TENDER / TYPICAL	R0 / 26-OCT-26	20 OF 30

BAND n50 CROSS-POL MACRO ANTENNA & RRU MOUNTING ASSEMBLY

+30.0M AGL • TRI-SECTOR PLATFORM • 76MM MOUNT PIPES • TILT ADJUSTABLE • RRU BRACKETS

TRI-SECTOR ANTENNA MOUNT ELEVATION & PLAN (SCALE 1:20)



ANTENNA & RRU MOUNTING SPECIFICATIONS

1. MOUNTING PLATFORM STRUCTURAL SPECIFICATIONS:

COMPONENT	DIMENSIONS / SPECIFICATION	MATERIAL
Collar Clamping Ring	450mm ID Heavy Split Collar	ASTM A572 Gr. 50 (12mm thk) / HDG
Outrigger Standoff Arms	75mm x 75mm x 6.0mm SHS	ASTM A500 Gr. B / HDG >= 85um
Antenna Mount Pipes	76mm OD (2.5-inch Nom) x 2.0m	Schedule 40 Galvanized Pipe
Mechanical Tilt Brackets	0 to 10 deg Down-tilt range	Heavy-duty pressed steel HDG
RRU Dedicated Pipe Mounts	60mm OD x 1.2m Galvanized Pipe	Mounted directly behind sector antenna
RF Coaxial Jumpers	1/2-inch Superflexible Heliax	DIN 4.3-10 weather-sealed connectors

2. SECTOR ORIENTATION & AZIMUTH SCHEDULE:

SECTOR	AZIMUTH	ANTENNA MODEL	TILT (MECH/ELEC)	RRU MODEL
Sector 1 (Alpha)	0.0 Degrees (True North)	Band n50 Cross-Pol 4T4R/8T8R	2 deg Mech / 4 deg Elec	8T8R 320W RRU
Sector 2 (Beta)	120.0 Degrees	Band n50 Cross-Pol 4T4R/8T8R	2 deg Mech / 4 deg Elec	8T8R 320W RRU
Sector 3 (Gamma)	240.0 Degrees	Band n50 Cross-Pol 4T4R/8T8R	2 deg Mech / 4 deg Elec	8T8R 320W RRU

3. ANTENNA RIGGING MANDATES:

- All RF connectors must be wrapped with 3 layers of rubber splicing tape followed by UV vinyl tape.
- RRU grounding pigtails (16mm2 green/yellow) must bond directly to the tower earthing busbar.
- Torque-seal tamper-evident inspection paint must be applied across all bracket clamping bolts.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

SHEET TITLE: BAND n50 CROSS-POL MACRO ANTENNA MOUNTING BRACKE

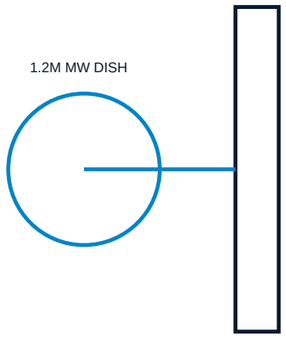
DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_ST_017	FOR TENDER / TYPICAL	R0 / 26-OCT-26	21 OF 30

STARLINK BACKHAUL & FUTURE MICROWAVE APPURTENANCE MOUNTS

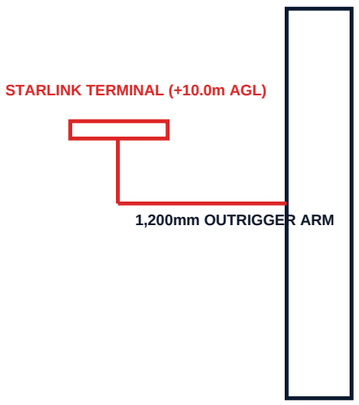
STARLINK @ +10M AGL • FUTURE 1.2M MW DISH @ +25M AGL • OUTRIGGER CANTILEVER • VIBRATION DAMPING

APPURTENANCE MOUNTING ELEVATIONS (SCALE 1:25)

1.2M SOLID MICROWAVE DISH MOUNT (+25.0M AGL):



STARLINK SATELLITE TERMINAL OUTRIGGER (+10.0M AGL):



SATELLITE & BACKHAUL MOUNTING SPECIFICATIONS

1. STARLINK SATELLITE TERMINAL SPECIFICATIONS:

PARAMETER	SPECIFICATION	OPERATIONAL DETAILS
Terminal Model	Starlink High-Performance Flat Panel	Electronic phased-array antenna
Operating Elevation	+10.00 Meters AGL	Clears 2.9m security fence and tree line
Cantilever Outrigger	1,200mm Cantilever Pipe Arm	60mm dia. Schedule 40 HDG Pipe
Sky View Clearance	100 deg Unobstructed Cone	Zero structural shadowing from tower tip
Cable Routing	Outdoor UV Cat6A Shielded	Runs internally within monopole shaft
Power Supply	100-240V AC / 56V DC PoE Injector	Powered from dedicated inverter / DC bus

2. FUTURE 1.2M MICROWAVE DISH SPECIFICATIONS:

PARAMETER	SPECIFICATION	DESIGN IMPACT
Antenna Diameter	1.20 Meters Solid Parabolic Dish	Included in 315 km/h wind reactions
Operating Elevation	+25.00 Meters AGL	Collar mount clamped on Section S5
EPA Drag Area	1.25 Sq. Meters (with Radome)	Wind force = 7.78 kN @ 315 km/h
Fine Azimuth Adjustment	+/- 5.0 Degrees Vernier Mount	Maintains sharp beam alignment
Strut Tie-back	1x 48mm dia. Diagonal Strut	Eliminates antenna torsional twist

3. BACKHAUL RIGGING MANDATES:

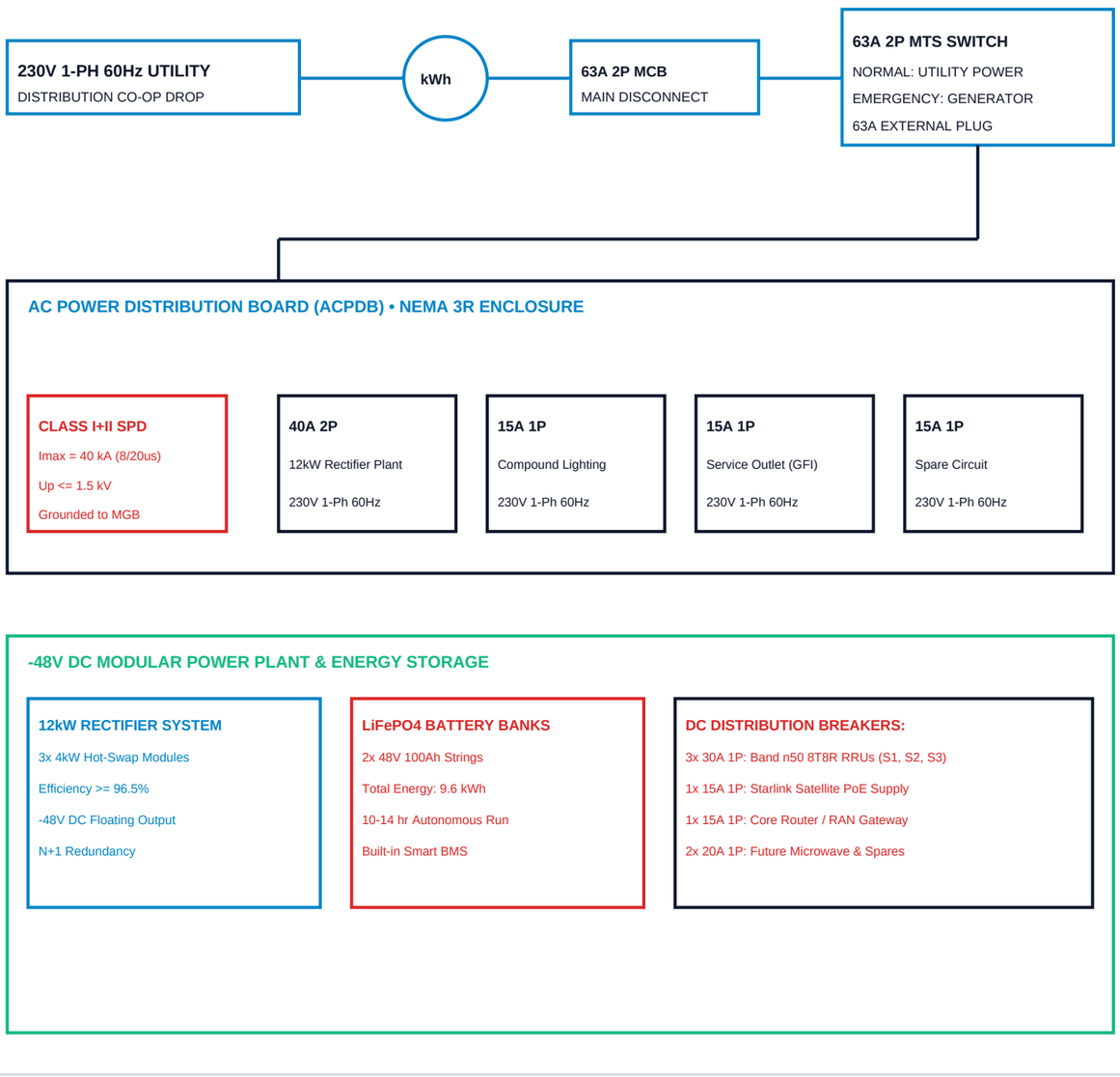
- Starlink terminal outrigger must point toward the northern sky (0 deg to 45 deg azimuth in PH latitude).
- Ethernet cable must incorporate an outdoor Gigabit Ethernet surge protector grounded to tower MGB.
- Future microwave mount capacity is certified under baseline tender wind loading with zero reinforcement required.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC.			TELTECH 5G FWA
PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50) SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa			
LEAD TECHNICAL ARCHITECT / REP: OLIVER TUNGOL	PROFESSIONAL STRUCTURAL ENGINEER: JOMAR DOGILLO, PE		
SHEET TITLE: STARLINK SATELLITE (10M AGL) & FUTURE EXPANSION			
DRAWING NO: TT_ST_018	RELEASE STATUS: FOR TENDER / TYPICAL	REV / DATE: R0 / 26-OCT-26	SHEET NO: 22 OF 30

ELECTRICAL SINGLE LINE DIAGRAM (SLD) & HYBRID POWER DISTRIBUTION

230V 1-PH UTILITY • 63A MTS • 12KW MODULAR RECTIFIER • -48V DC BUSBAR • LIFEPO4 BATTERIES

ELECTRICAL SINGLE LINE DIAGRAM (SLD) SCHEMATIC (NTS)



ELECTRICAL LOAD SCHEDULE & SPECIFICATIONS

1. CONNECTED LOAD SCHEDULE & CURRENT DRAW:

LOAD DESCRIPTION	VOLTAGE	WATTS	AMPS	CB RATING
12kW Rectifier Full Load	230V AC	4,800 W	20.9 A	40A 2P
Perimeter Security Lights	230V AC	100 W	0.4 A	15A 1P
Service Utility Outlet (GFI)	230V AC	1,000 W	4.3 A	15A 1P
TOTAL AC CONNECTED LOAD	230V 1-Ph	5,900 W	25.6 A	63A Main
Band n50 8T8R RRUs (3x)	-48V DC	2,850 W	59.4 A	3x 30A 1P
Starlink Satellite PoE	-48V DC	150 W	3.1 A	15A 1P
RAN Gateway / Switch	-48V DC	120 W	2.5 A	15A 1P
TOTAL DC TELECOM LOAD	-48V DC	3,120 W	65.0 A	100A Bus

2. SURGE PROTECTION & BATTERY BACKUP SPECIFICATIONS:

PROTECTION ELEMENT	SPECIFICATION	COMPLIANCE STANDARD
AC Surge Protective Device	Class I+II (Imax = 40kA, Up <= 1.5kV)	IEC 61643-11 / UL 1449
DC Surge Suppression	Class II Din-rail (-48V DC, 20kA)	IEC 61643-21
Battery Chemistry	Lithium Iron Phosphate (LiFePO4)	UL 1973 / IEC 62619
Battery Operating Life	>= 4,000 Cycles @ 80% DoD	10+ Year Design Life
Generator Transfer Switch	63A 2-Pole Break-Before-Make	UL 1008 / PEC Section 7.0

3. ELECTRICAL INSTALLATION MANDATES:

- All wiring shall strictly comply with the Philippine Electrical Code (PEC 2017 Part 1).
- Conductors shall be 99.9% oxygen-free high-conductivity annealed copper with THHN/THWN-2 insulation.
- Color coding: Phase A (Black/Red), Neutral (White), Ground (Continuous Green or Green with Yellow Stripe).

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

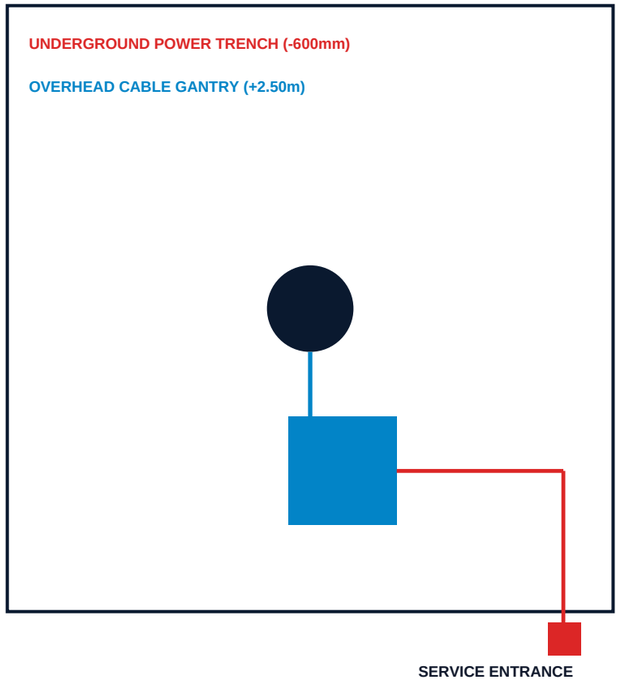
SHEET TITLE: ELECTRICAL SINGLE LINE DIAGRAM (SLD) & POWER DIS

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_EL_001	FOR TENDER / TYPICAL	R0 / 26-OCT-26	23 OF 30

SITE ELECTRICAL TRENCHING, CONDUIT ROUTING & POWER DUCTING

UNDERGROUND PVC CONDUITS • 600MM BURIAL DEPTH • CABLE SEPARATION • ROXTEC TRANSIT BOOT

ELECTRICAL TRENCHING & CONDUIT LAYOUT (SCALE 1:100)



CONDUIT & TRENCHING SPECIFICATIONS

1. CONDUIT ROUTING & DUCT SPECIFICATIONS:

ROUTE SEGMENT	CONDUIT SIZE / TYPE	BURIAL DEPTH	CONTENTS
Service Post to ACPDB	50mm dia. Schedule 40 PVC	600 mm Below Grade	3-14 mm2 THHN Copper Feeder
ACPDB to Tower Cable Tray	Overhead Gantry Bridge	+2.50m Clearance	DC Power & Fiber Feeders
ACPDB to Perimeter Lights	25mm dia. Schedule 40 PVC	450 mm Below Grade	3-3.5 mm2 THHN Lighting Feeds
MGB to Perimeter Earth Ring	50mm dia. Heavy PVC Sleeve	800 mm Below Grade	50 mm2 Bare Copper Leads
Backhaul Pull-Box to Plinth	110mm dia. HDPE Conduit	800 mm Below Grade	48-Core Fiber Optic Cable

2. TRENCH CROSS-SECTION & EMBEDMENT SPECIFICATIONS:

TRENCH LAYER	MATERIAL SPECIFICATION	THICKNESS / POSITION
Surface Reinstatement	Gravel surfacing or native soil	Top 150mm depth
Backfill Material	Clean granular soil, compacted 95%	150mm to 500mm depth
Warning Ribbon Tape	Bright Yellow Polyethylene Tape	300mm Below Finished Grade
Concrete Protective Cap	75mm Lean Concrete Envelope	Directly overlying conduits
Conduit Sand Bedding	Clean natural sand bedding	75mm below & 75mm above ducts

3. TRENCHING MANDATES:

- AC power conduits and fiber optic ducts must maintain 300mm minimum lateral clearance in shared trenches.
- Yellow warning ribbon marked 'CAUTION: BURIED ELECTRIC LINE' must be installed along entire trench run.
- All conduits entering equipment cabinets must be sealed with duct sealant putty to prevent moisture & pest entry.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — **TELTECH 5G FWA**

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

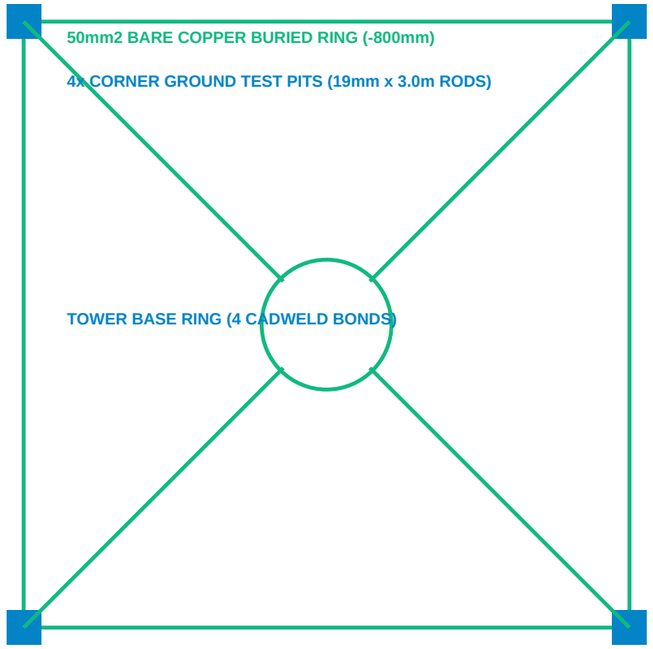
SHEET TITLE: SITE ELECTRICAL LAYOUT, TRENCHING & CONDUIT ROUT

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_EL_002	FOR TENDER / TYPICAL	R0 / 26-OCT-26	24 OF 30

COMPREHENSIVE SITE EARTHING & GROUNDING GRID MASTER PLAN

CLOSED RING GRID • 50MM2 BARE COPPER • 4 CORNER TEST PITS • CADWELD WELDS • R <= 5.0 OHMS

SITE GROUNDING GRID PLAN (SCALE 1:100)



GROUNDING SYSTEM TECHNICAL SPECIFICATIONS

1. GROUNDING GRID MATERIAL SPECIFICATIONS:

COMPONENT	MATERIAL / RATING	SPECIFICATION / CODE
Perimeter Ring Conductor	50 mm2 Bare Stranded Copper	Soft-drawn electrolytic copper (ASTM B8)
Vertical Ground Rods	19mm dia. x 3,000mm Length	Copper-clad steel (minimum 254um copper)
Tower Bonding Leads	50 mm2 Bare Stranded Copper	4 Diagonal leads connected via Cadweld
Exothermic Welds	Cadweld Molecular Fusion	IEEE 837 permanent molecular bond
Soil Enhancement Material	Bentonite / Conductive Carbon	Applied around rods in high-resistivity soil
Main Grounding Busbar (MGB)	500mm x 100mm x 6mm Copper	Tinned electrolytic copper (99.9% Cu)

2. RESISTANCE TO GROUND MANDATES & LIMITS:

TEST PARAMETER	MAXIMUM ALLOWABLE	TELTECH DESIGN GOAL
Combined Site Earth Resistance	R <= 5.00 Ohms	Target R <= 3.00 Ohms
Lightning Down-Conductor Path	R <= 1.00 Ohm	Direct low-impedance path
Fence Equipotential Bonding	R <= 0.10 Ohm (Continuity)	Zero touch-potential hazard
Measurement Methodology	3-Point Fall-of-Potential	IEEE 81 standard testing protocol

3. EARTHING EXECUTION MANDATES:

- Ground ring conductor must be buried at minimum 800mm depth below finished grade to prevent disturbance.
- All underground conductor-to-conductor and conductor-to-rod joints must be executed using Cadweld exothermic welds.
- Mechanical compression clamps are strictly forbidden in underground buried grounding networks.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

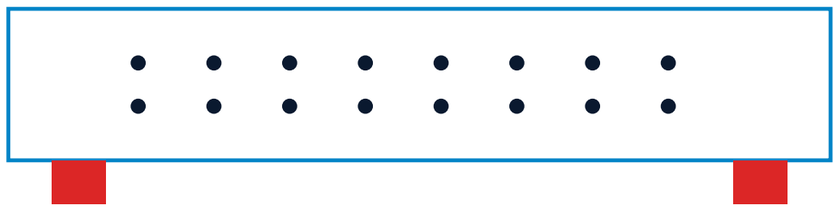
SHEET TITLE: SITE EARTHING & GROUNDING GRID LAYOUT PLAN (<= 5

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_EL_003	FOR TENDER / TYPICAL	R0 / 26-OCT-26	25 OF 30

MAIN GROUNDING BUSBAR (MGB) & EQUIPOTENTIAL TERMINATIONS

500x100x6MM TINNED COPPER • TWO-HOLE NEMA SPACING • 600V INSULATORS • LABELED LUGS

MAIN GROUNDING BUSBAR (MGB) ELEVATION (SCALE 1:10)



MGB CONNECTION SCHEDULE & TERMINATION DETAILS

1. MGB TERMINATION ALLOCATION MATRIX:

POS	CONNECTED SUBSYSTEM	CONDUCTOR SIZE	LUG TYPE / HARDWARE
P1	Tower Lightning Down-Conductor	70 mm2 Bare Copper	2-Hole Long Barrel Copper Lug
P2	Tower Base Shell Grounding	50 mm2 Bare Copper	2-Hole Long Barrel Copper Lug
P3	AC Power SPD Ground Lead	35 mm2 Insulated Green	2-Hole Long Barrel Copper Lug
P4	-48V DC Power Plant Reference	50 mm2 Insulated Green	2-Hole Long Barrel Copper Lug
P5	Outdoor Equipment Plinth Rebar	50 mm2 Bare Copper	2-Hole Long Barrel Copper Lug
P6	Overhead Cable Gantry Bridge	25mm x 3mm Copper Tape	2-Hole Heavy Copper Lug
P7	Perimeter Fence Bonding Lead	50 mm2 Bare Copper	2-Hole Long Barrel Copper Lug
P8	Buried Earth Grid Ring Connection	50 mm2 Bare Copper	2-Hole Long Barrel Copper Lug

2. HARDWARE & TORQUE SPECIFICATIONS:

BOLT SIZE	MATERIAL	SPECIFIED TORQUE	CORROSION COMPOUND
3/8-16 UNC	Grade 316 Stainless Steel	25 Nm (18 ft-lbs)	Antioxidant Joint Compound
1/2-13 UNC	Grade 316 Stainless Steel	55 Nm (40 ft-lbs)	Antioxidant Joint Compound

3. TERMINATION DIRECTIVES:

- All connections must use two-hole compression lugs with standard NEMA hole spacing (19mm to 44.5mm).
- Busbar contact surfaces must be cleaned to bright metal and coated with antioxidant compound (Penetrox).
- Every connection lug must be permanently labeled with an engraved phenolic tag identifying the subsystem.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

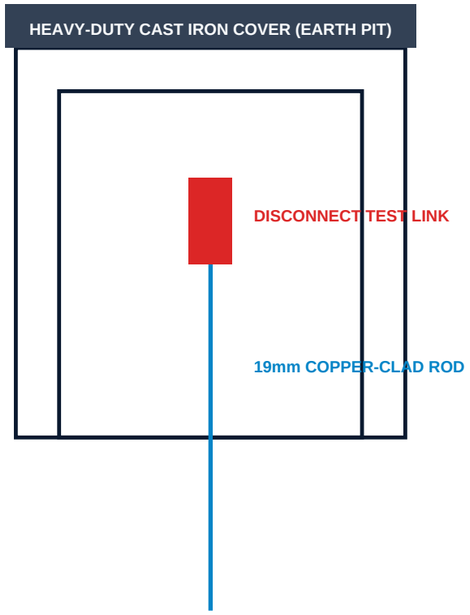
SHEET TITLE: MAIN GROUNDING BUSBAR (MGB) & EARTH PIT CONNECTI

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_EL_004	FOR TENDER / TYPICAL	R0 / 26-OCT-26	26 OF 30

GROUND TEST PIT & INSPECTION CHAMBER MASTER DETAILS

300x300x300MM PRECAST CONCRETE • CAST IRON COVER • REMOVABLE COPPER TEST LINK

EARTH TEST PIT SECTION & PLAN (SCALE 1:15)



TEST PIT SPECIFICATIONS & TESTING PROTOCOL

1. TEST PIT STRUCTURAL SPECIFICATIONS:

COMPONENT	DIMENSION / RATING	SPECIFICATION
Chamber Dimensions	300mm W x 300mm L x 300mm D	Precast reinforced concrete (f'c >= 28 MPa)
Inspection Cover	Class B125 Cast Iron Cover	Marked 'EARTH TEST PIT' / Traffic-rated
Ground Rod Extension	+150 mm Above Pit Bottom	Allows unencumbered clamp attachment
Disconnect Test Link	50mm x 6mm Solid Copper Link	Removable via stainless steel bolts
Gravel Drainage Bed	100mm Coarse Crushed Stone	Provides internal stormwater drainage

2. ANNUAL TESTING PROTOCOL (FALL-OF-POTENTIAL):

TEST PROCEDURE	CRITERIA / REQUIREMENT	ACTION IF FAILED
Disconnect Test Link	Isolate rod under test	Verify zero stray voltage before testing
Current Probe (C2)	Place probe at 30m distance	Direct line from test pit
Potential Probe (P2)	Place probe at 62% distance (18.6m)	Record resistance value
Resistance Benchmark	Individual Rod: R <= 25 Ohms	Apply soil enhancement compound if R > 25
System Combined Grid	All Links Closed: R <= 5.0 Ohms	Install additional deep ground rods if R > 5

3. TEST PIT DIRECTIVES:

- Covers must sit flush with compound gravel surfacing to eliminate tripping hazards for site personnel.
- Disconnect links must never be removed while the telecommunications site is energized under storm conditions.
- After annual resistance measurement, all disconnect links must be torqued tight and coated with protective grease.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — **TELTECH 5G FWA**

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

SHEET TITLE: GROUND TEST PIT & INSPECTION CHAMBER DETAILS

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_EL_005	FOR TENDER / TYPICAL	R0 / 26-OCT-26	27 OF 30

LIGHTNING PROTECTION SYSTEM (LPS) & CADWELD WELD DETAILS

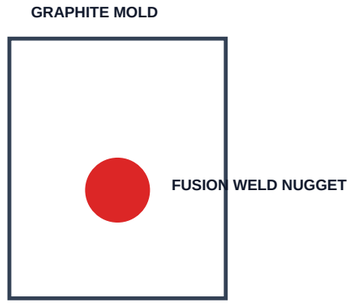
+31.5M AGL LIGHTNING ROD • 70MM2 DOWN-CONDUCTOR • 4x CADWELD BASE FUSION BONDS

LPS DETAILS & CADWELD MOLDS (SCALE 1:20)

16MM DIA. x 1.5M LIGHTNING ROD AT TOWER PEAK (+31.5M AGL):



CADWELD EXOTHERMIC MOLECULAR FUSION BOND:



LPS & EXOTHERMIC WELDING SPECIFICATIONS

1. LIGHTNING PROTECTION SYSTEM (NFPA 780 / IEC 62305):

LPS ELEMENT	SPECIFICATION	COMPLIANCE MANDATE
Air Terminal Tip	16mm dia. x 1,500mm Stainless/Copper	Extends +1.5m above highest antenna
Zone of Protection	Rolling Sphere Radius R = 45m	100% protective cone covers all RF gear
Main Down-Conductor	70 mm2 Bare Stranded Copper	Runs internally down monopole tube
Tower Base Bonds	4x Cadweld Connections	Symmetrically bonded to buried earth ring
Surge Current Rating	100 kA (10/350 us waveform)	Direct lightning strike withstand rating

2. CADWELD EXOTHERMIC WELD PROCESS PARAMETERS:

JOINT CONFIGURATION	MOLD TYPE	WELD METAL	PULL STRENGTH
Cable to Steel Pole Base	Cadweld Type VN / VS	F20 or #90 Cartridge	>= 100% Cable Yield
Cable to Ground Rod Top	Cadweld Type GR / GT	#90 or #115 Cartridge	>= 100% Cable Yield
Cable to Cable Splice	Cadweld Type SS / TA	#90 Cartridge	>= 100% Cable Yield

3. CADWELD QUALITY CONTROL DIRECTIVES:

- Surfaces to be welded must be ground clean to bare, shiny metal free of moisture, zinc, paint, and rust.
- Molds must be pre-heated with propane torch prior to the first weld of the day to eliminate moisture.
- Finished welds must be dense, solid, free of slag inclusions, and withstand a sharp blow from a 1-lb hammer.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

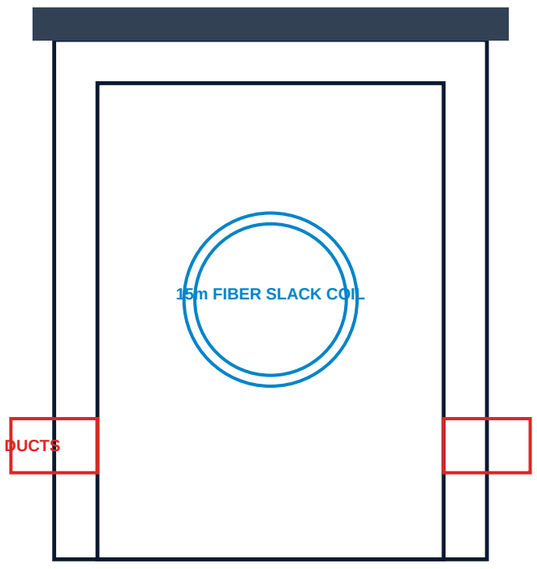
SHEET TITLE: TOWER LIGHTNING DOWN-CONDUCTOR & CADWELD EXOTHER

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_EL_006	FOR TENDER / TYPICAL	R0 / 26-OCT-26	28 OF 30

FIBER OPTIC BACKHAUL LEAD-IN & UNDERGROUND PULL-BOX DETAILS

2-WAY 110MM HDPE DUCTS • 600x600x800MM PULL-BOX • 15M FIBER SLACK LOOP BRACKET

FIBER PULL-BOX SECTION & SLACK LOOP (SCALE 1:20)



FIBER BACKHAUL & DUCTING SPECIFICATIONS

1. FIBER INFRASTRUCTURE SPECIFICATIONS:

COMPONENT	DIMENSION / RATING	SPECIFICATION
Backhaul Ducts	2-Way 110mm OD SDR-11 HDPE	Thick-walled smooth interior with pull rope
Underground Pull-Box	600mm W x 600mm L x 800mm D	Precast reinforced concrete (f'c >= 28 MPa)
Access Cover	Galvanized Diamond Plate (6mm)	Traffic-rated with flush padlock clasps
Fiber Slack Coil	15.0 Meters Service Slack	Stored on galvanized steel wall-mount brackets
Optical Fiber Cable	48-Core Single-Mode (G.652D)	Armored loose-tube outdoor cable
Optical Distribution Frame	1RU 19-inch Rack Mount ODF	24-Port SC/APC with splice tray in cabinet

2. DUCT INSTALLATION & TESTING DIRECTIVES:

TEST / MANDATE	CRITERIA	PURPOSE / DETAILS
Mandrel Test	85% Internal Diameter Mandrel	Verifies zero duct ovality or crushing
Pressure Testing	Pneumatic 1.0 Bar for 15 min	Ensures watertight joints along entire run
Duct Sealing Plugs	Mechanical expandable rubber plugs	Seals empty conduits against silt & rodent entry

3. FIBER SPLICING MANDATES:

- Fiber optic cables must maintain minimum bending radius >= 20 times outer diameter under tension.
- Fusion splice insertion loss shall not exceed 0.05 dB per splice at 1310nm and 1550nm wavelengths.
- OTDR bidirectional trace reports must be archived in final tender as-built handover documentation.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. — TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

SHEET TITLE: FIBER OPTIC LEAD-IN, DUCTING & UNDERGROUND PULL-

DRAWING NO:	RELEASE STATUS:	REV / DATE:	SHEET NO:
TT_EL_008	FOR TENDER / TYPICAL	R0 / 26-OCT-26	29 OF 30

GEOTECHNICAL SOIL BEARING CAPACITY & REMEDIATION PROTOCOLS

BASELINE qa >= 150 KPA • SPT N60 >= 15 • SOFT SOIL MITIGATION (100-150 KPA) • BOREHOLE PROTOCOL

GEOTECHNICAL COMPLIANCE CRITERIA & REMEDIATION DECISION MATRIX

1. GEOTECHNICAL BASELINE REQUIREMENTS (qa >= 150 kPa):

GEOTECHNICAL PARAMETER	BASELINE REQUIREMENT	VERIFICATION PROTOCOL
Allowable Soil Bearing (qa)	qa >= 150.0 kPa (15.3 ton/m2)	Standard Penetration Test (SPT) / Triaxial
Standard Penetration (N60)	N60 >= 15 Blows / 300mm	Continuous SPT sampling at 1.5m intervals
Founding Depth (-1.70m)	Competent undisturbed stratum	Visual verification by Geotechnical Engineer
Groundwater Table (GWT)	Assumed at -1.0m (Conservative)	Hydrostatic buoyant uplift checked
Exploratory Boreholes	Two (2) Boreholes to 10.0m Depth	ASTM D1586 rotary drilling with SPT

2. SITE CLASSIFICATION & SPT CORRELATION (NSCP 2015 TABLE 208-1):

SOIL PROFILE TYPE	SPT BLOWS (N60)	UNDRAINED SHEAR (Su)	qa RANGE
Class A (Hard Rock)	N > 100	Su > 250 kPa	qa >= 500 kPa
Class B (Rock / Dense)	50 <= N <= 100	100 <= Su <= 250 kPa	qa = 300 - 500 kPa
Class C (Very Dense / Stiff)	15 <= N <= 50	50 <= Su <= 100 kPa	qa = 150 - 300 kPa (BASELINE)
Class D (Stiff / Medium)	10 <= N <= 15	25 <= Su <= 50 kPa	qa = 100 - 150 kPa (REMEDIATION)
Class E (Soft Clay)	N < 10	Su < 25 kPa	qa < 100 kPa (SPECIAL PILES)

3. SOFT SOIL REMEDIATION ACTION MATRIX (qa < 150 kPa down to 100 kPa):

METHOD	ENGINEERING SOLUTION	APPLICABILITY / COST
OPTION A: Enlarged Mat Footing	Expand mat to 6.0m x 6.0m x 0.9m thick (q_max = 82 kPa <= 100 kPa)	Preferred if site boundaries permit
OPTION B: Engineered Rock Mattress	Over-excavate 500mm soft soil; place compacted crusher-run gravel / boulders	High efficacy for high water table
OPTION C: Bored Micro-Piles	Install 4x 250mm dia. steel-cased micro-piles socketed to competent rock	Mandatory if soft clay exceeds 3m depth

4. GEOTECHNICAL SIGN-OFF & MANDATORY CERTIFICATION:

- A site-specific geotechnical soil report signed by a licensed Geotechnical Engineer is mandatory for each site.
- If field SPT yields N60 < 15 at founding elevation, work must halt immediately and trigger the Remediation Matrix.
- All backfill over the footing must be placed in 200mm lifts and compacted to 95% Modified Proctor density.
- Foundation excavation inspection sign-off sheet must form part of the final LGU building permit submission.

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INC. TELTECH 5G FWA

PROJECT: NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50)
SPEC: 30M MONOPOLE • 315 KPH (NSCP 2015 ZONE 1) • qa = 150 kPa

LEAD TECHNICAL ARCHITECT / REP:	PROFESSIONAL STRUCTURAL ENGINEER:
OLIVER TUNGOL	JOMAR DOGILLO, PE

SHEET TITLE: GEOTECHNICAL SOIL INVESTIGATION & BEARING CAPACI

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